

模拟电子技术基础

Analog Circuits

1.2 Diode 二极管

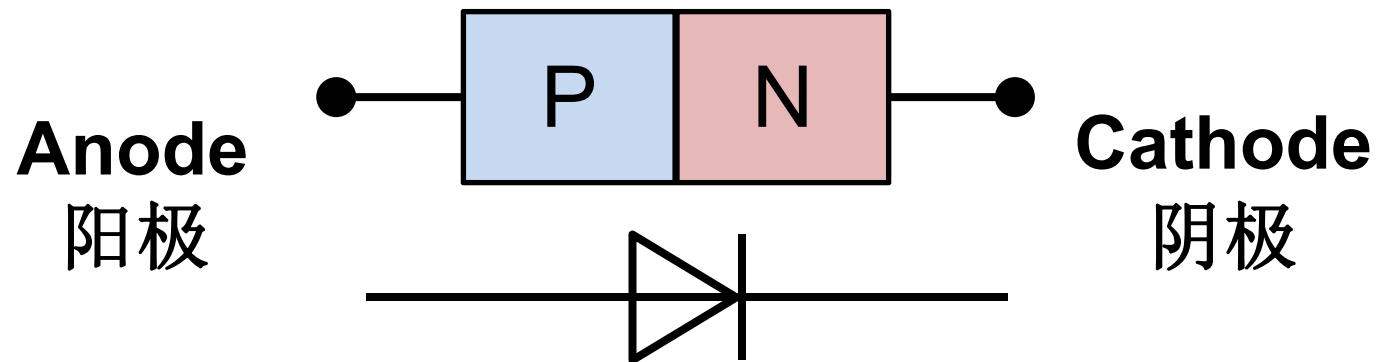
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Department of Electrical and Electronic Engineering



1.2 Diode 二极管

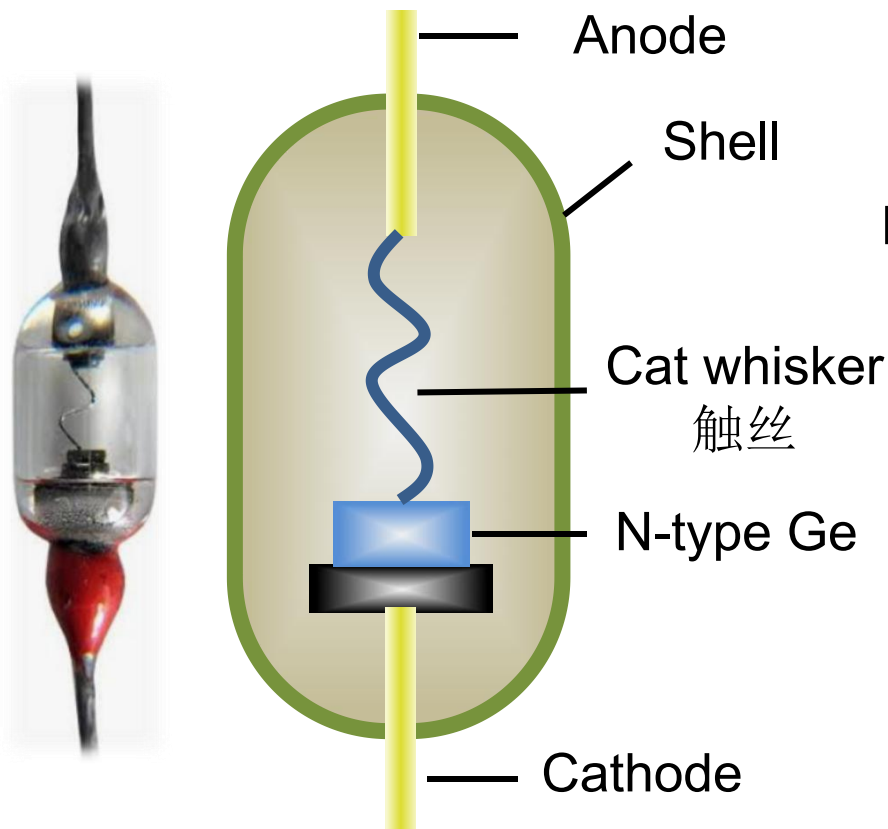
Diode: connect PN junction with wires and seal with shell.



1.2.1 Types of diode

Point contact diode

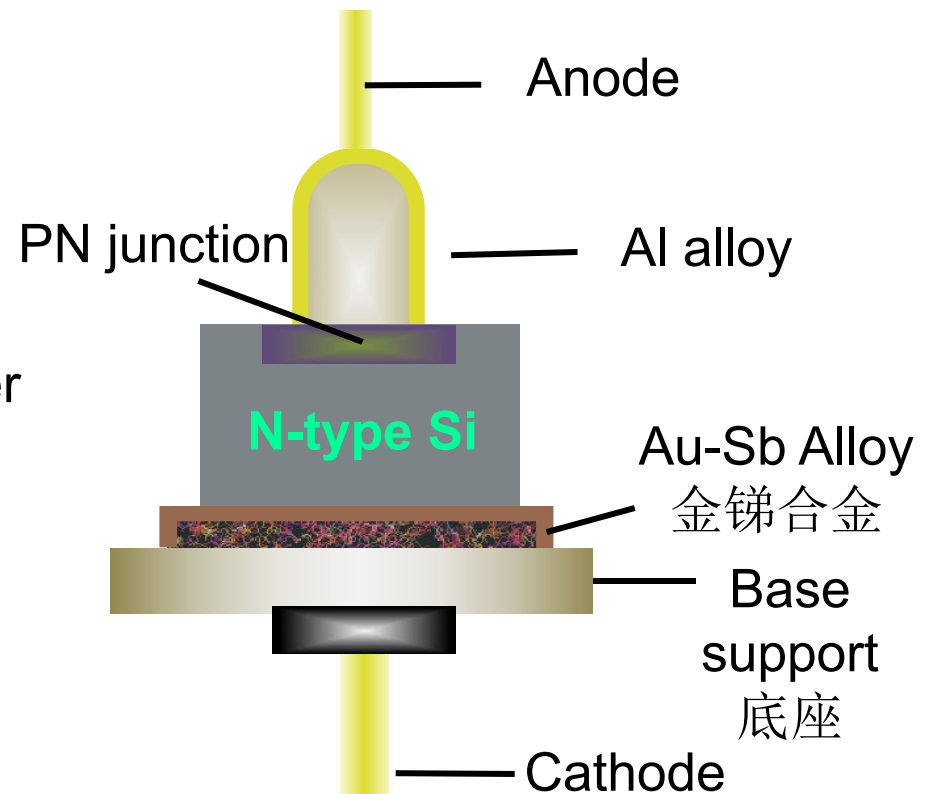
点接触型



Small contact area: high frequency

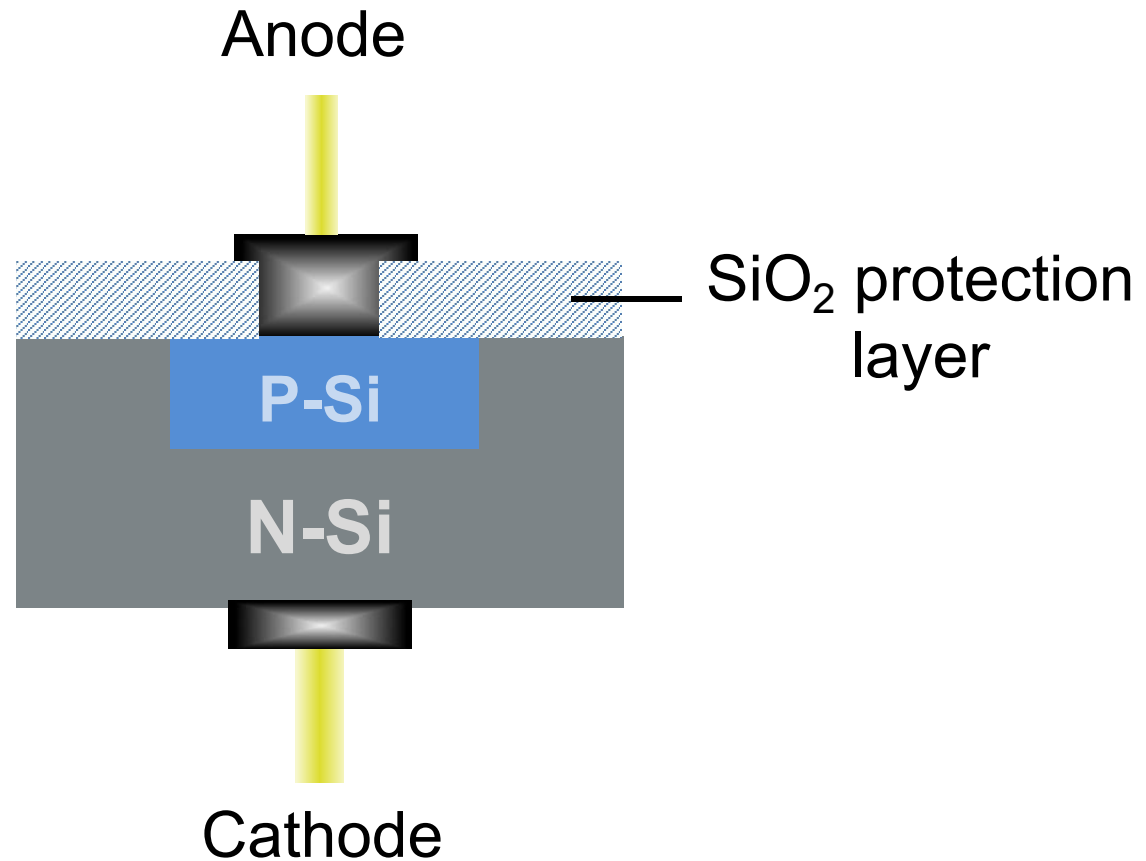
Surface contact diode

面接触型



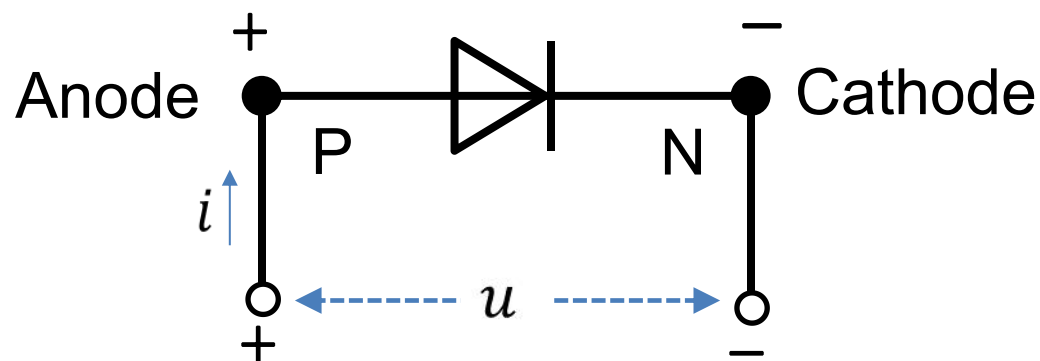
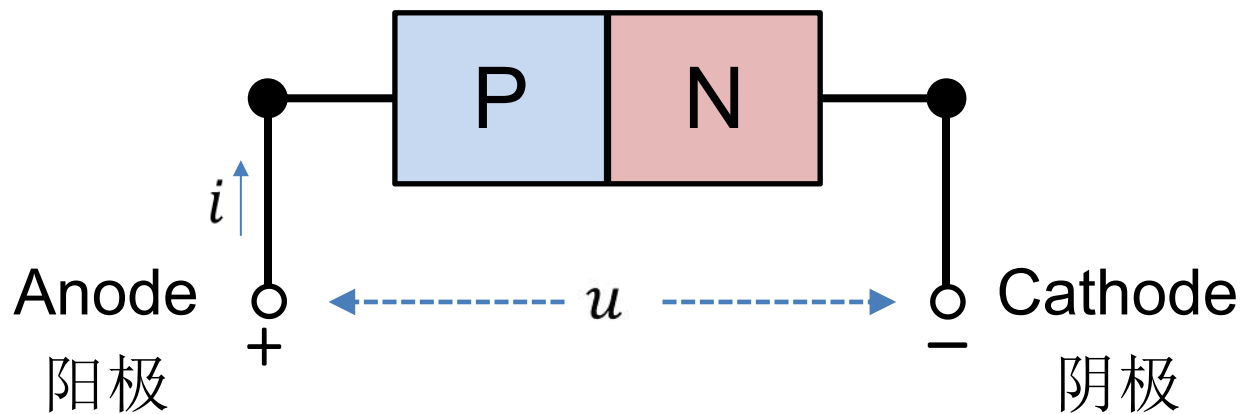
Large contact area: high current applications

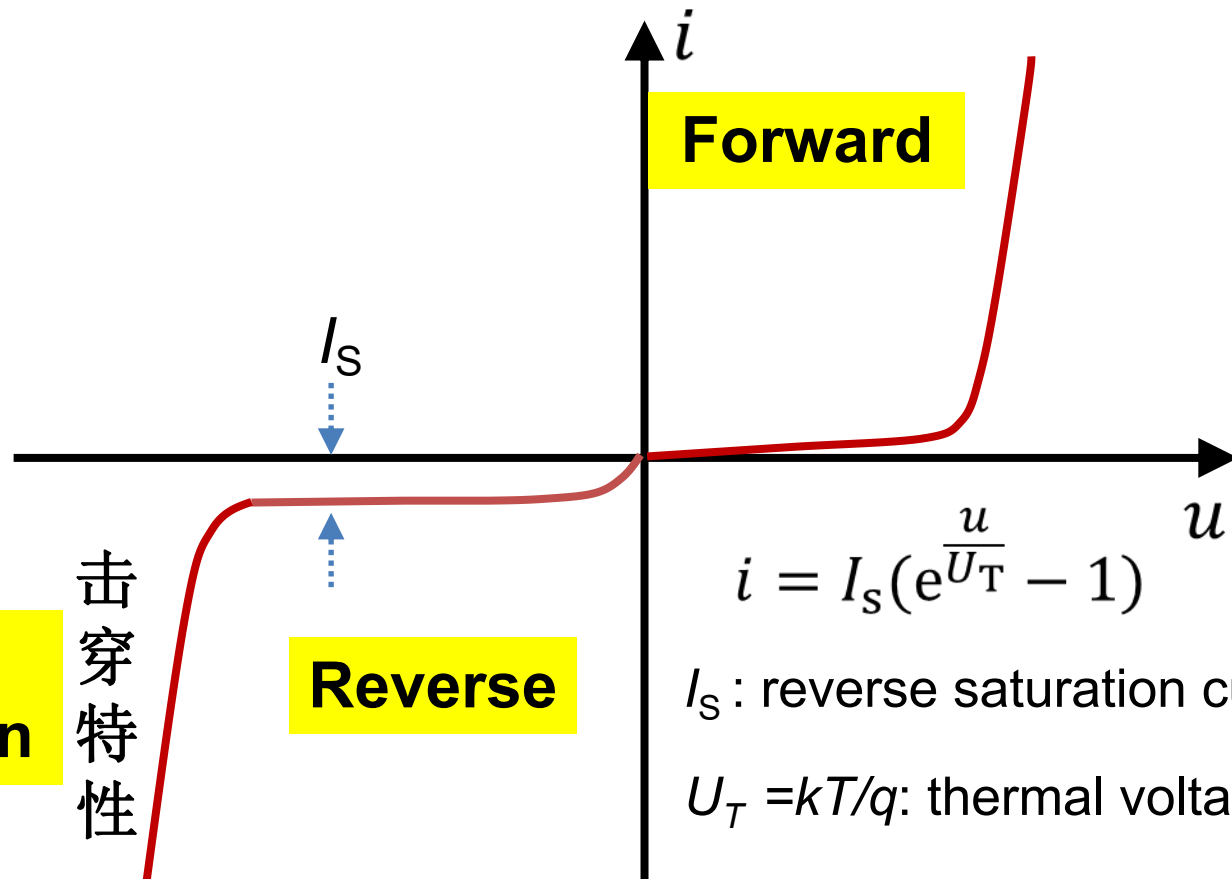
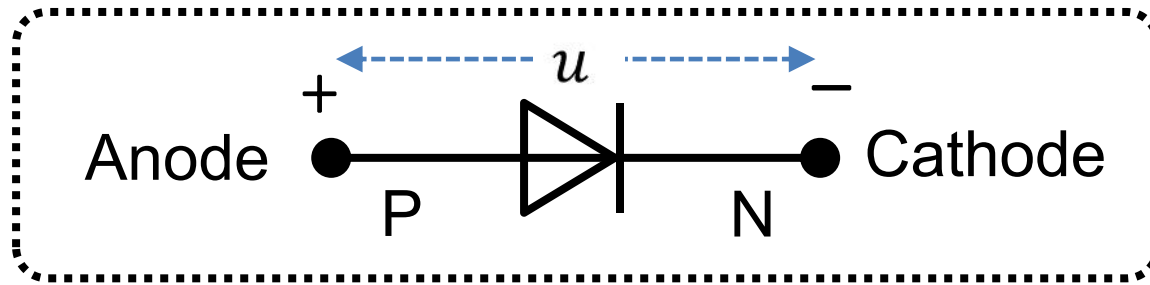
Planar diode平面型



Good for integrated circuit!

1.2.2 Volt-ampere characteristics 伏安特性





Forward

Reverse

**Reverse
Breakdown**

击穿特性

$$i = I_s (e^{\frac{u}{U_T}} - 1)$$

I_s : reverse saturation current

$U_T = kT/q$: thermal voltage

$T=300K, U_T=26 \text{ mV}$

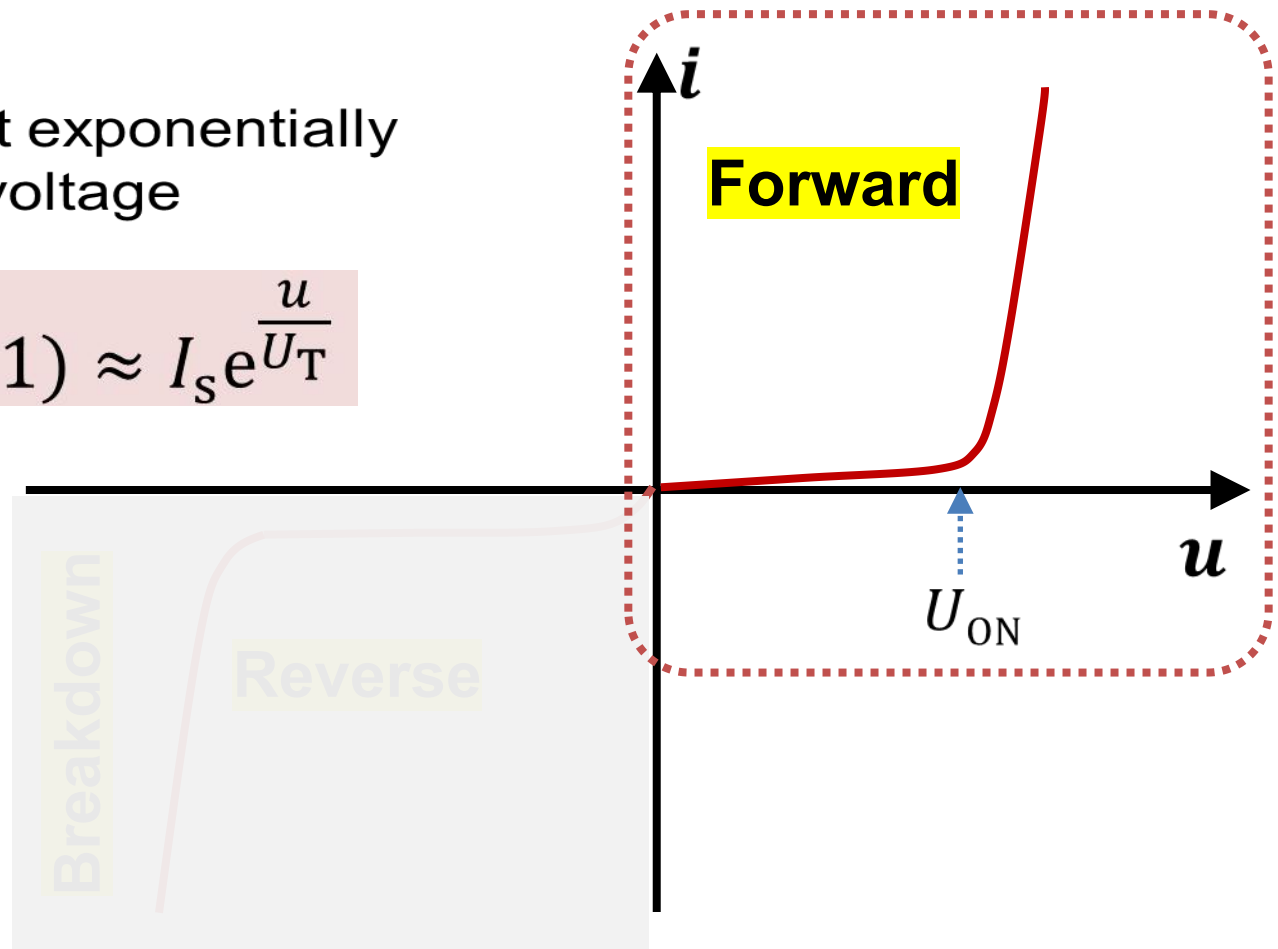
Forward characteristic 正向特性

When $u > U_{\text{ON}}$, diode is on.

U_{ON} : cut-in voltage 开启电压, $\approx$ Barrier potential of PN junction

$u > U_{\text{ON}}$, current exponentially increases with voltage

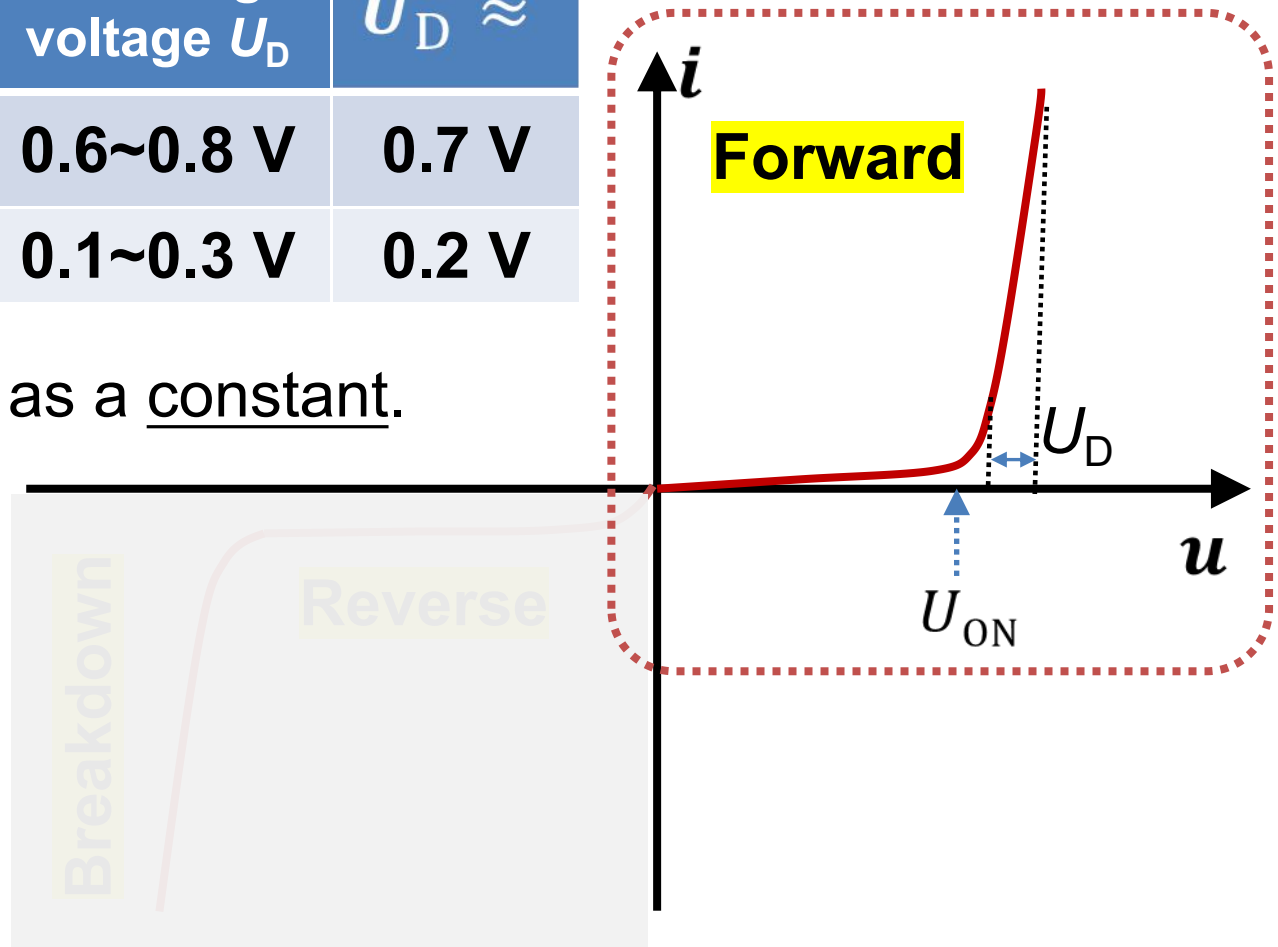
$$i = I_s \left(e^{\frac{u}{U_T}} - 1 \right) \approx I_s e^{\frac{u}{U_T}}$$



Forward characteristic

Diode	Cut-in voltage U_{ON}	Working voltage U_D	$U_D \approx$
Si	$\approx 0.5 \text{ V}$	$0.6 \sim 0.8 \text{ V}$	0.7 V
Ge	$\approx 0.1 \text{ V}$	$0.1 \sim 0.3 \text{ V}$	0.2 V

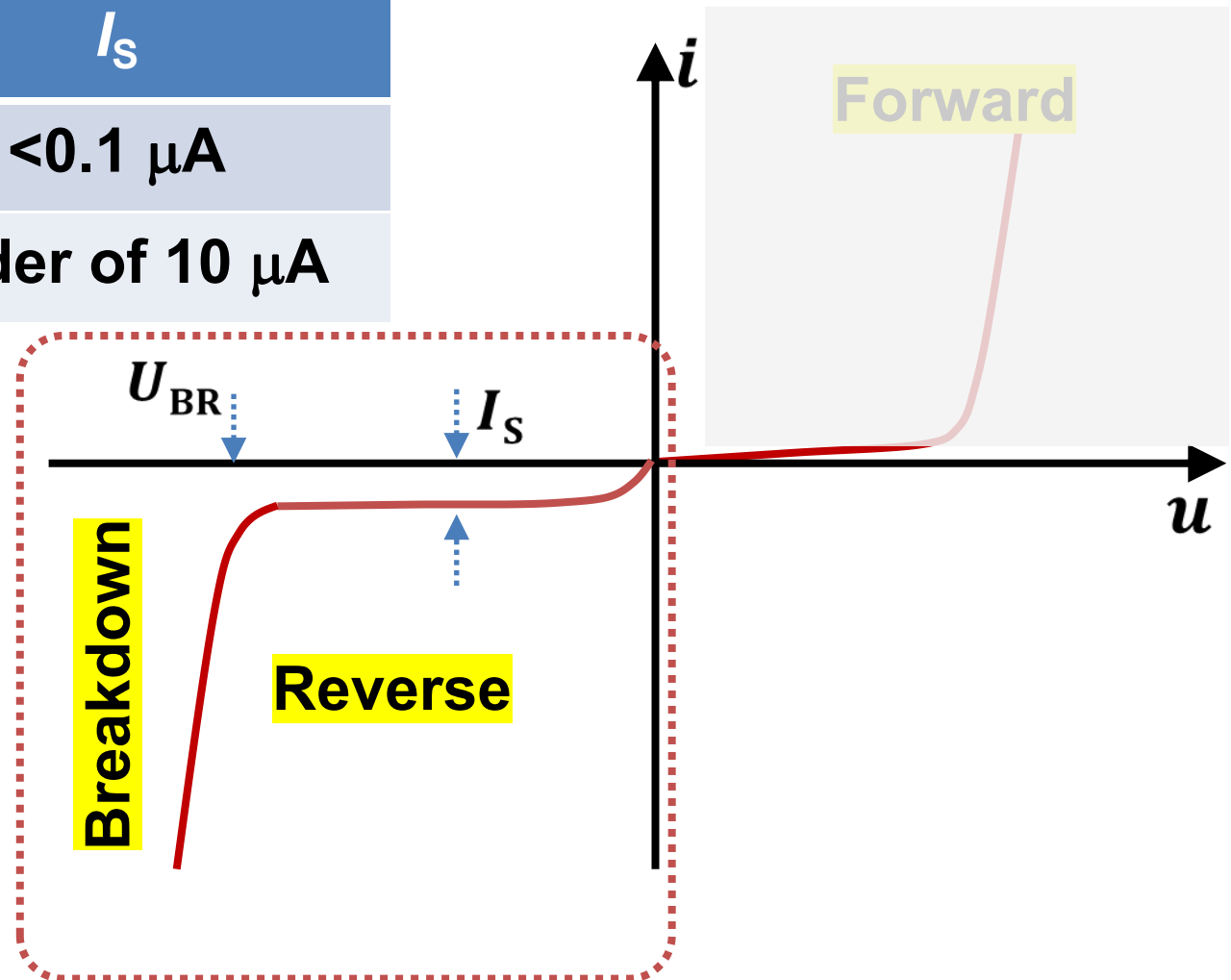
U_D can be treated as a constant.



Reverse characteristic 反向特性

$|u| < |U_{BR}|$, $i \approx I_S$, Reverse saturation

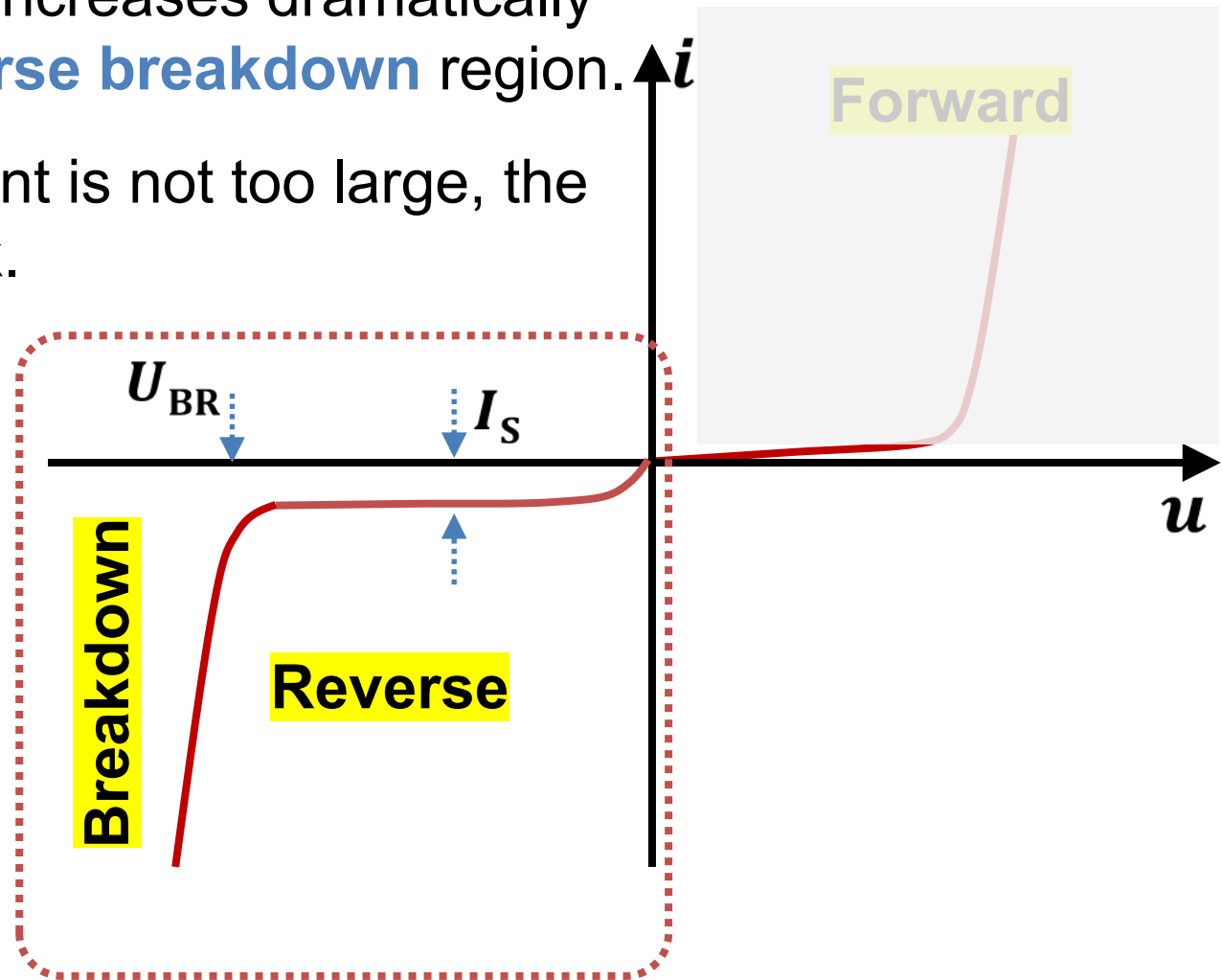
Diode	I_S
Si	$< 0.1 \mu A$
Ge	Order of $10 \mu A$



Reverse breakdown

$|u| > |U_{BR}|$, current increases dramatically and enter the **reverse breakdown** region.

If the reverse current is not too large, the diode can still work.



Reverse breakdown

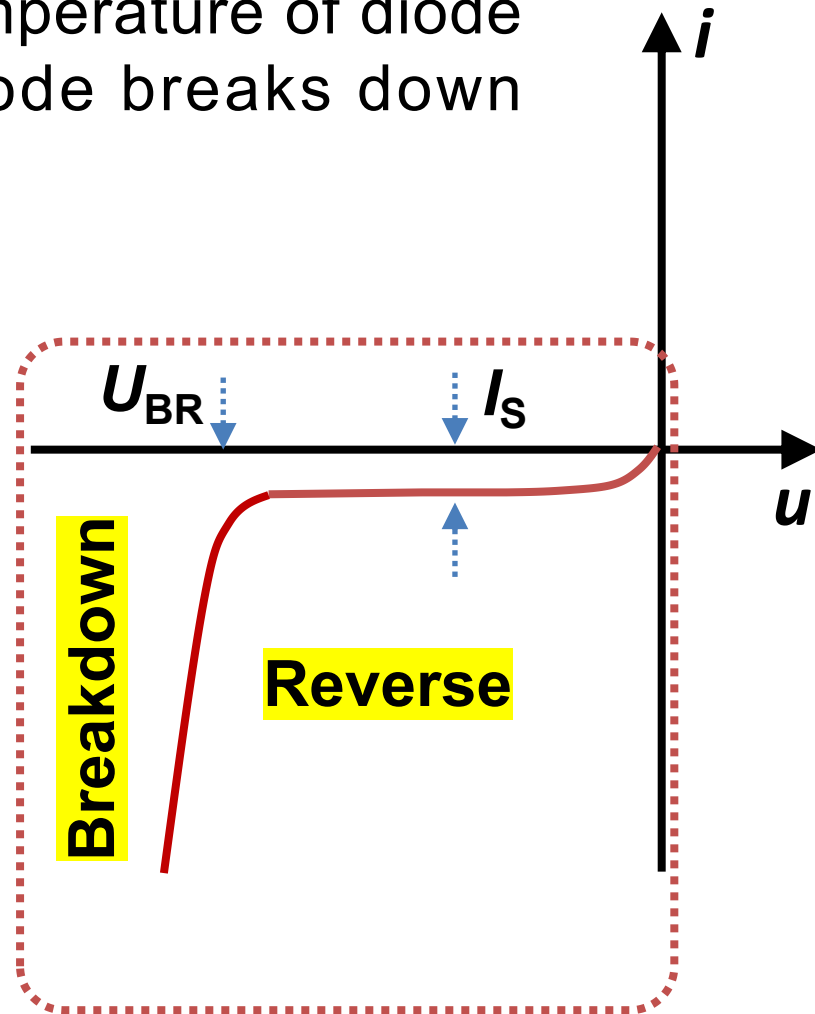
If the current is too large, the temperature of diode increases significantly, and diode breaks down permanently.

Diode	Breakdown temperature
Si	150~200 °C
Ge	75~100 °C

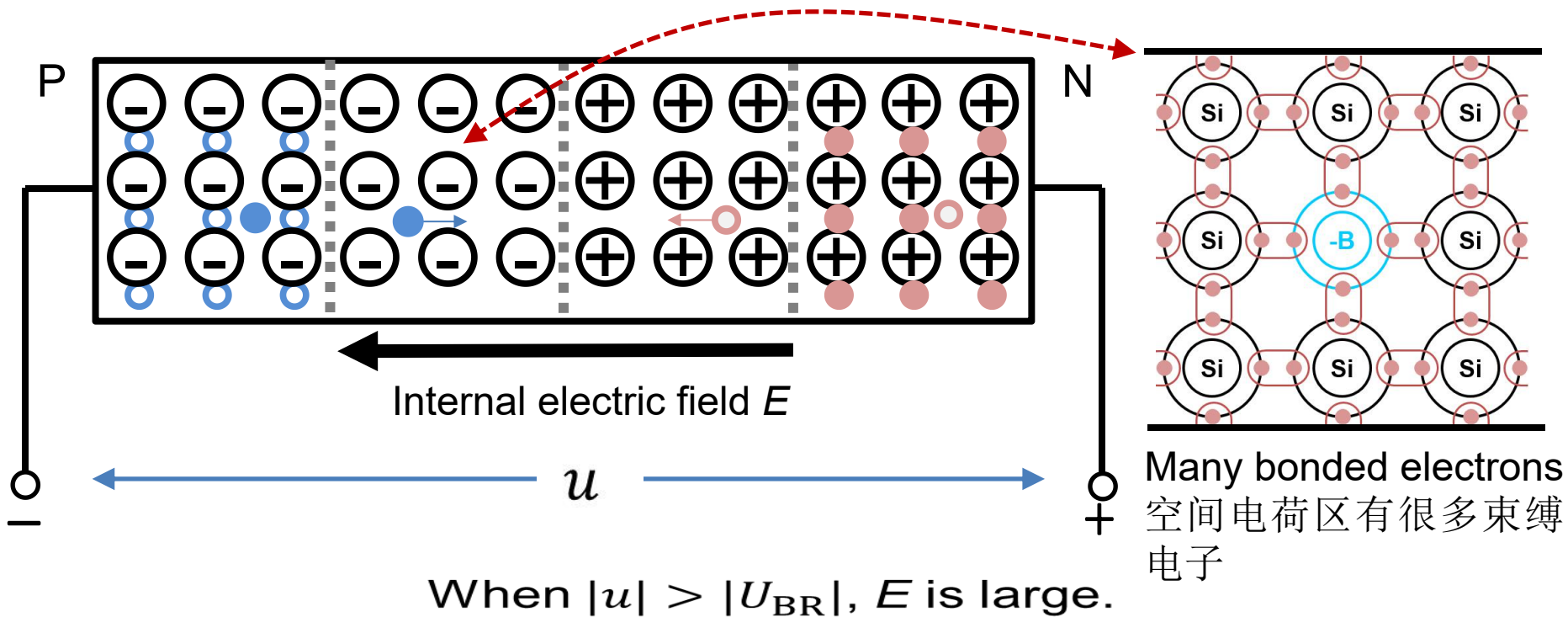
Breakdown mechanism:

Avalanche breakdown 雪崩击穿

Zener breakdown 齐纳击穿

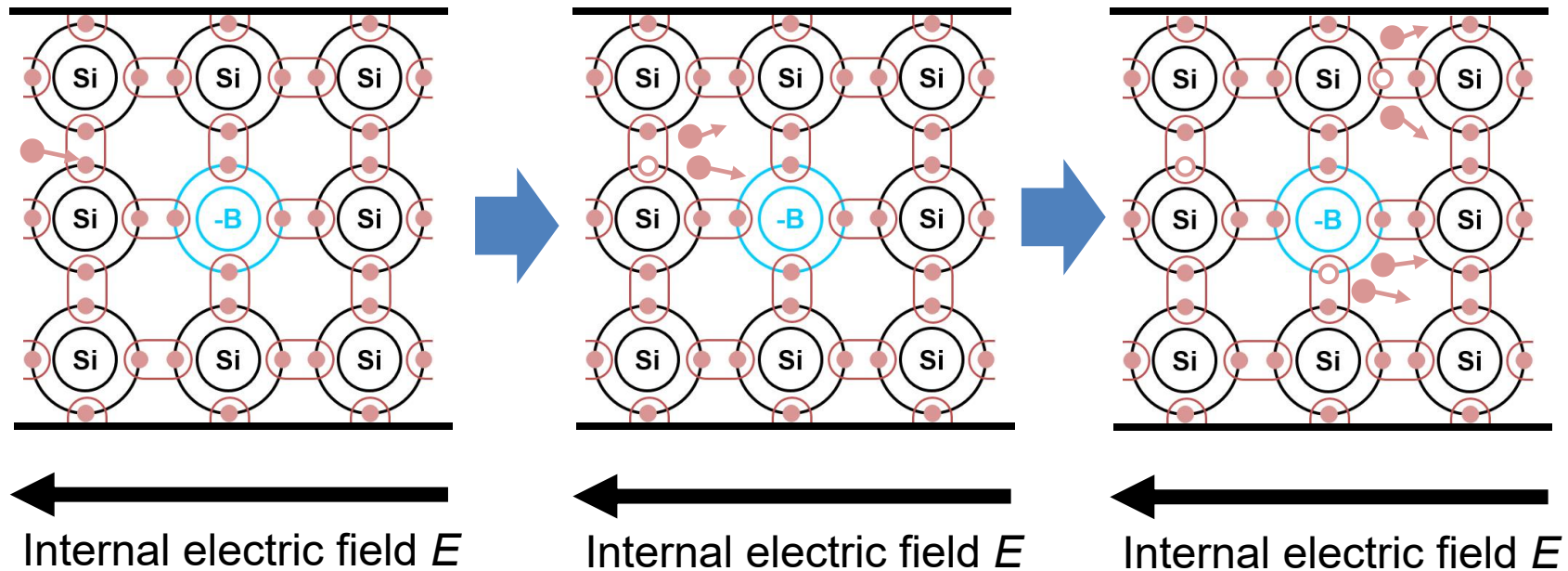


Avalanche breakdown 雪崩击穿



Minor carriers are accelerated to a very high speed and energy.

Avalanche breakdown 雪崩击穿

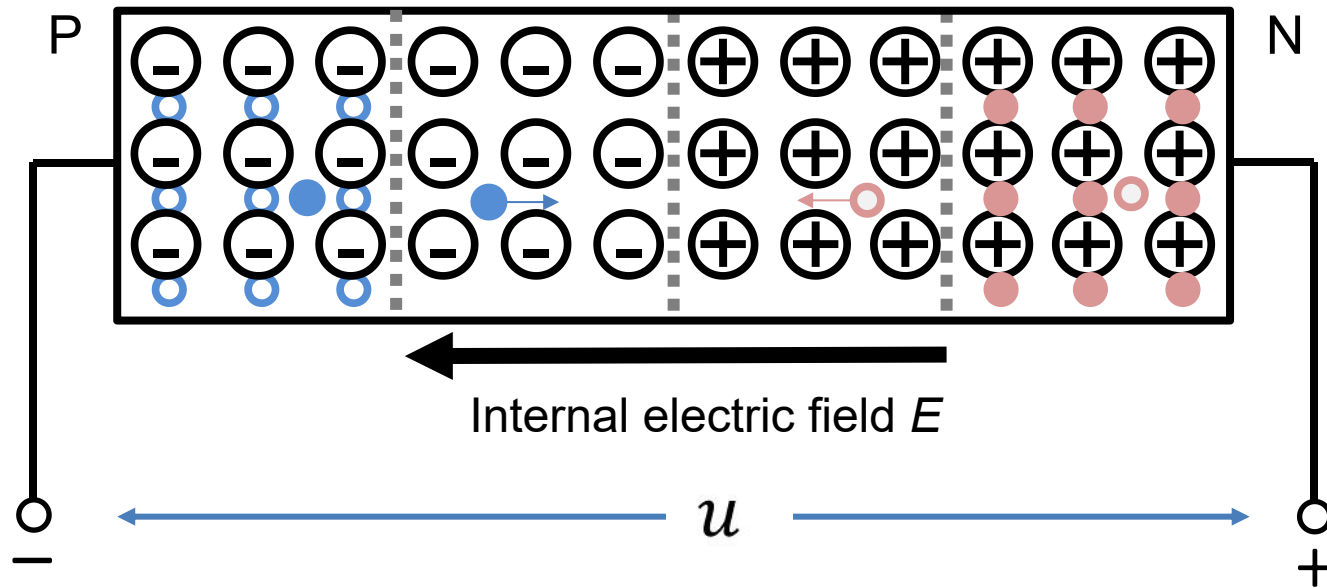


Accelerated electrons collide with bonded electrons and kick them out.



More and more carriers are generated and current increases dramatically!

Avalanche breakdown 雪崩击穿



The condition for observing the avalanche breakdown:

- Depletion region is wide.
- Low doping concentration.

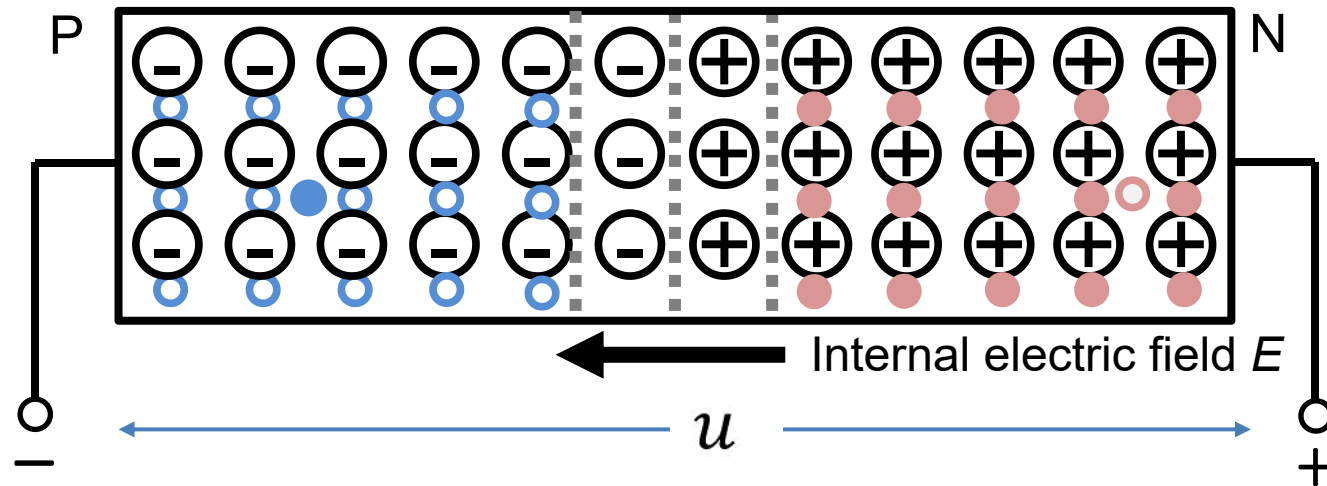
雪崩击穿 **Avalanche breakdown**

条件 { 半导体的掺杂浓度低
耗尽层宽，击穿电压相对高

通过空间电荷区的电子和空穴，会在电场作用下，使获得的能量增大。在晶体中运行的电子和空穴将不断的与晶体原子发生碰撞，通过这样的碰撞可使束缚在共价键中的价电子碰撞出来，产生自由电子-空穴对。新产生的载流子在电场作用下撞出其他价电子，又产生新的自由电子和空穴对。如此连锁反应，使得阻挡层中的载流子的数量雪崩式地增加，流过PN结的电流就急剧增大击穿PN结，这种碰撞电离导致击穿称为雪崩击穿。

击穿的特点 { 击穿电压高于6V
击穿电压具有正的温度系数

Zener breakdown 齐纳击穿

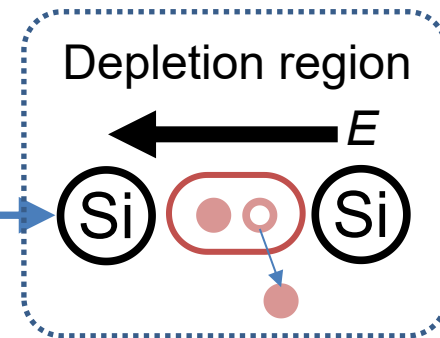


In heavily doped PN junction, the depletion region is **narrow**.

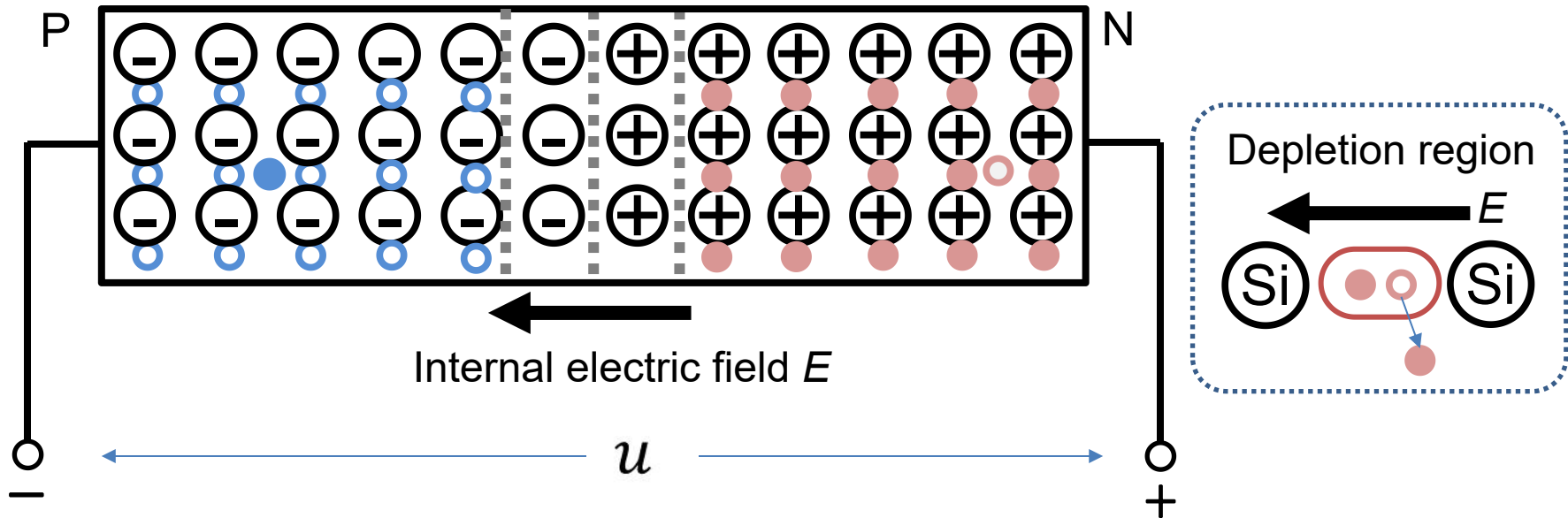
When u increases, E increases.

Bonded electrons can be directly **pulled out** by the strong electric field.

Large amount of free electrons and holes are generated and current increases dramatically!



Zener breakdown 齐纳击穿



Force: electric field

Carrot: bonded electrons

Other explanation: quantum tunneling 量子隧穿

齐纳击穿 Zener breakdown

条件 { 半导体的掺杂浓度高
耗尽层窄，击穿电压相对低

当反向电压增大到一定程度时，空间电荷区内就会建立一个很强的电场。这个强电场能把价电子从共价键中拉出来，从而在空间电荷区产生大量电子-空穴对。这些电子-空穴对产生后，空穴被强电场驱到P区，电子被强电场驱到N区，使反向电流猛增。这种由于强电场的作用，直接产生大量电子-空穴对而使反向电流剧增的现象叫做**齐纳击穿**。

击穿的特点 { 击穿电压低于**4V**
击穿电压具有负的温度系数

Temperature effect on the volt-ampere characteristics

Temperature $\uparrow$

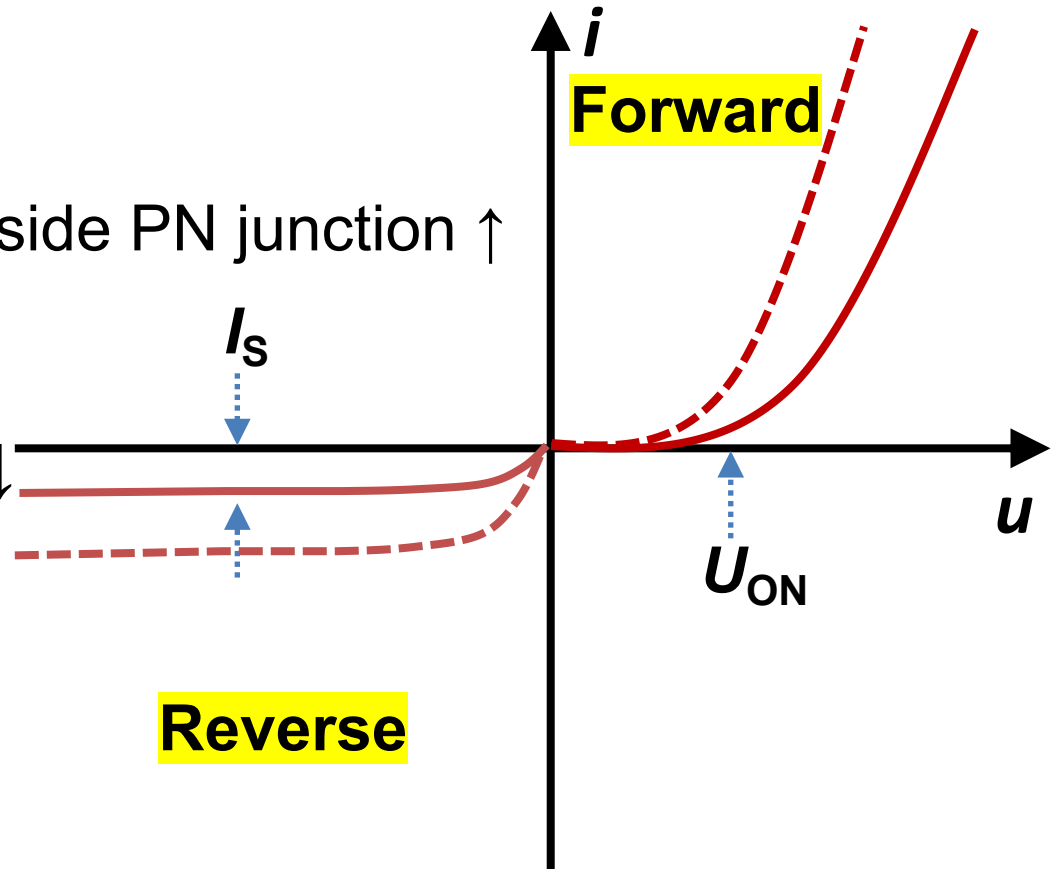


Free carriers inside/outside PN junction $\uparrow$



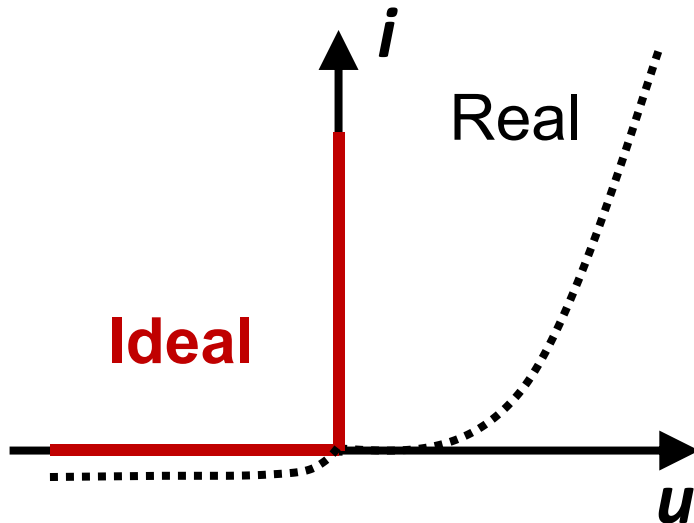
Threshold voltage $U_{ON} \downarrow$

Reverse current $I_S \uparrow$

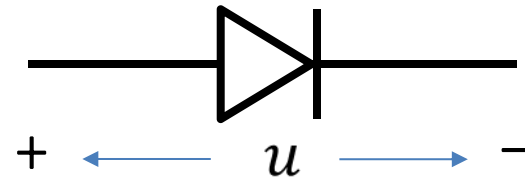


1.2.4 The model of diodes: forward

Ideal diode 理想二极管



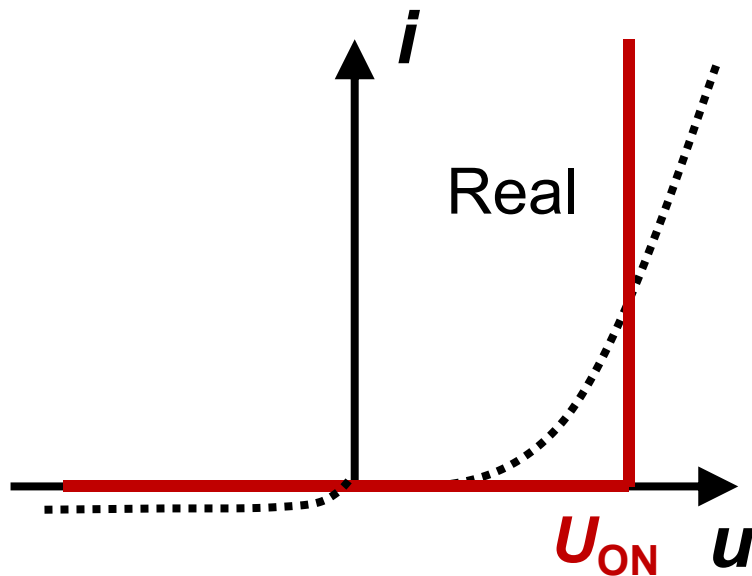
The symbol for ideal diode



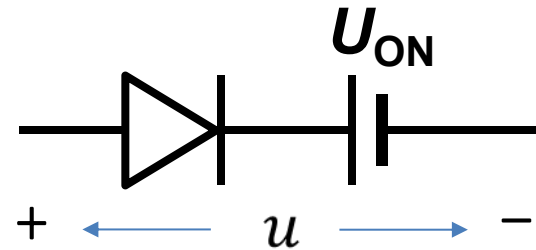
$$\begin{cases} u > 0: r = 0 \\ u \leq 0: r \rightarrow \infty \end{cases}$$

Constant-voltage-drop model / Offset model 恒压模型

Q: The symbol?

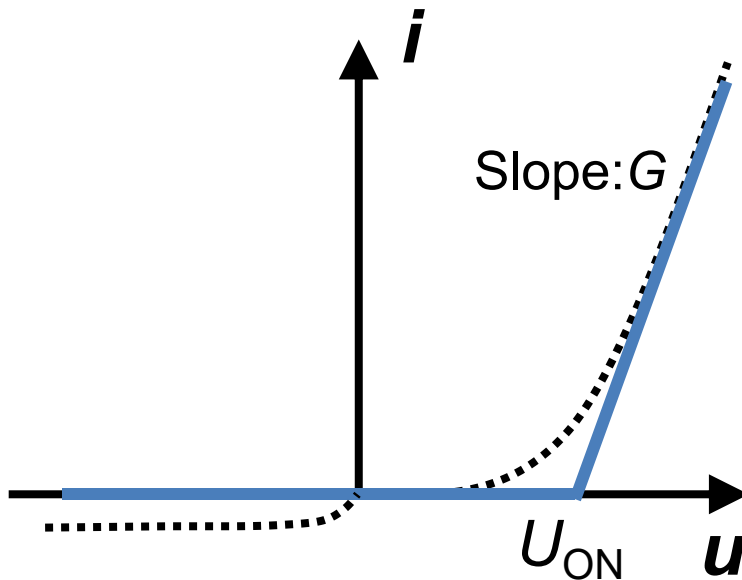


$$\begin{cases} u > U_{ON}: r = 0 \\ u \leq U_{ON}: r \rightarrow \infty \end{cases}$$



Offset-resistance model 折线模型

Q: What's the symbol?



$$u \leq U_{ON}:$$

$$r \rightarrow \infty$$

$$u > U_{ON}:$$

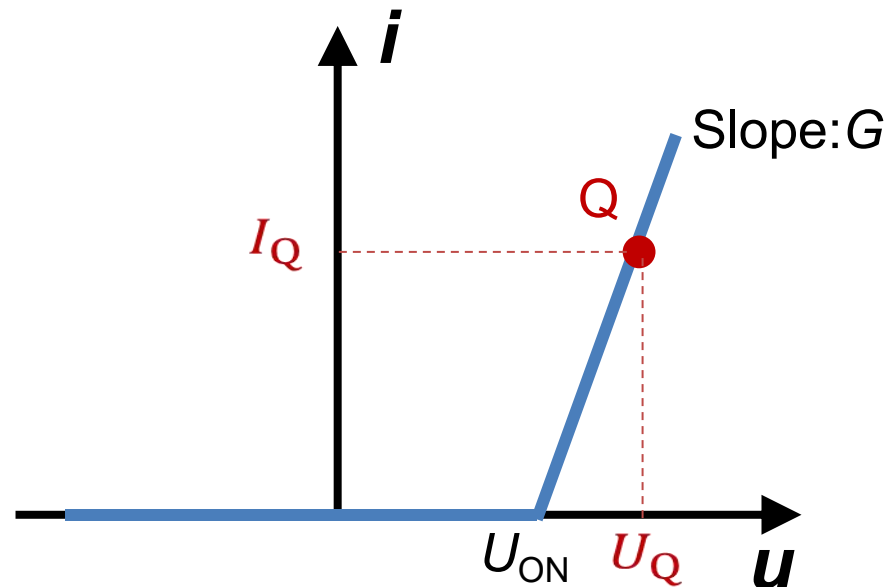
$$i = G(u - U_{ON})$$



$$u = U_{ON} + i/G$$



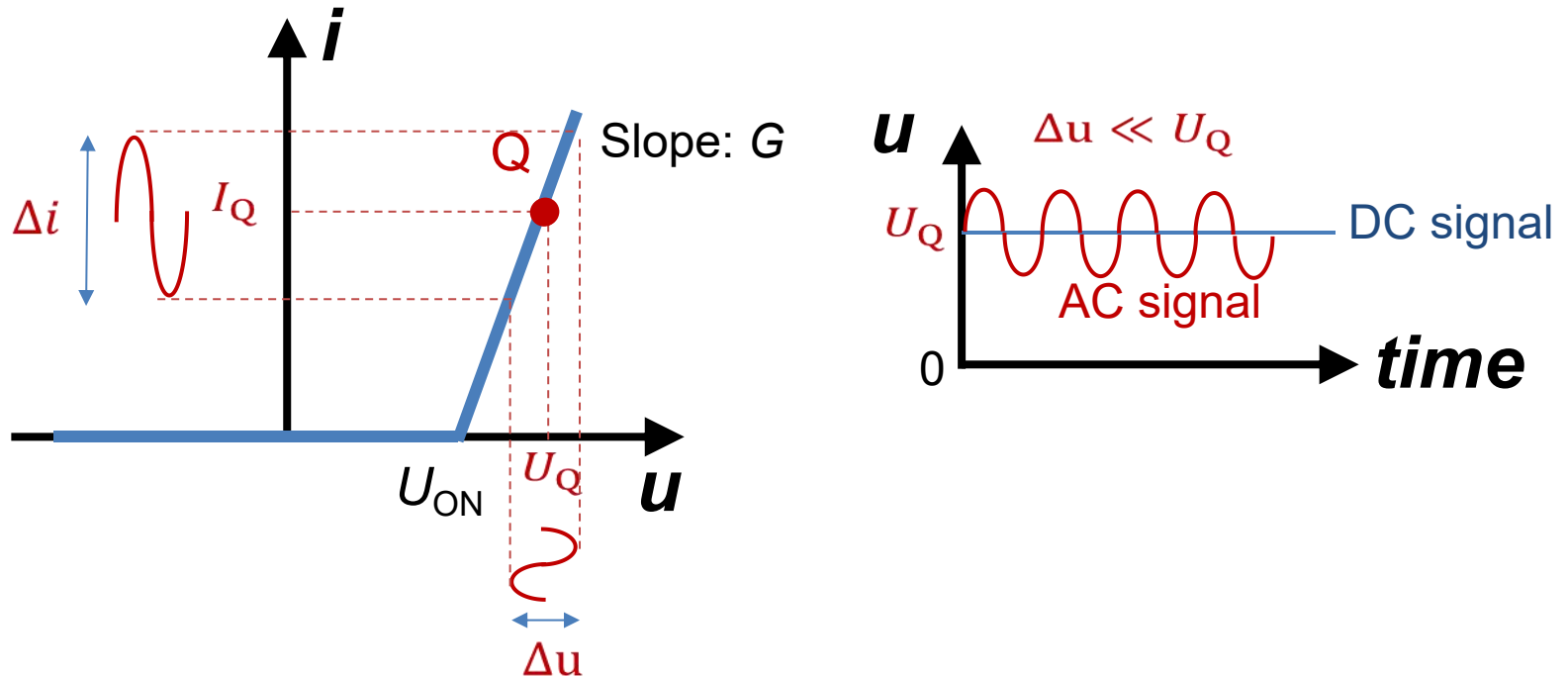
Question: what's the DC resistance at Q point?



DC resistance at voltage U_Q : $R_{DC} = U_Q / I_Q$

R_{DC} is not a constant as a function of voltage u

Question: what's the AC/dynamic resistance at Q point?

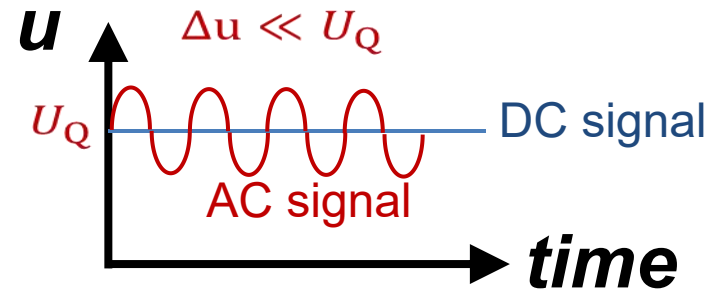
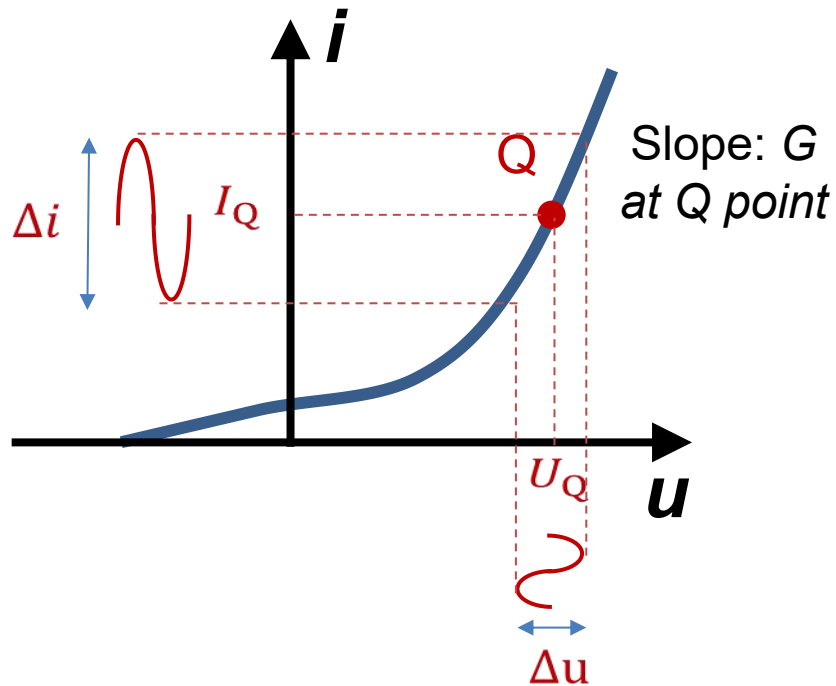


AC/dynamic resistance at Q: voltage/current signal is AC

$$R_{AC} = \left. \frac{du}{di} \right|_{U_Q} = \frac{1}{G} = r_D$$

R_{AC} is a **constant** as a function of voltage u

DC and AC resistance at Q point for an arbitrary curve



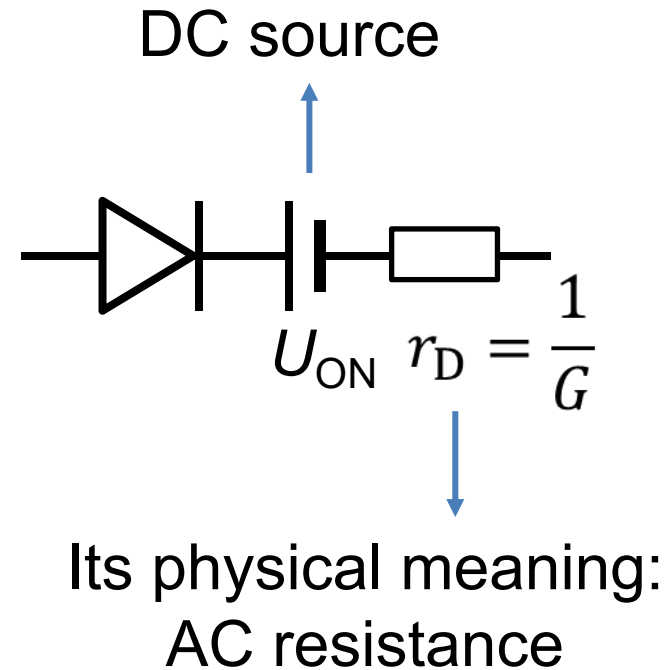
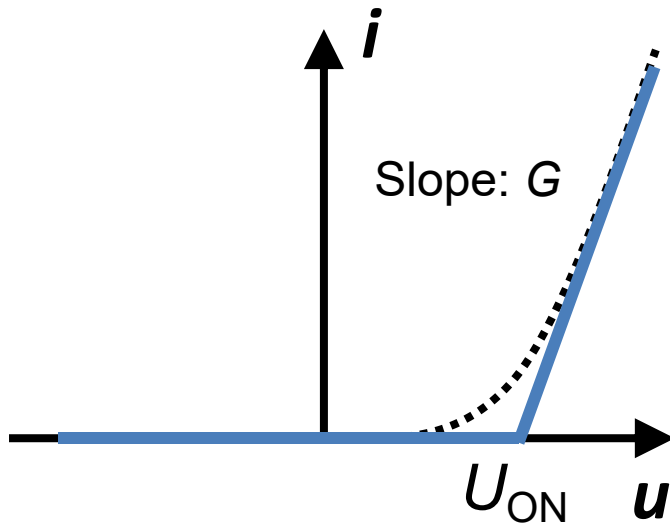
DC/static resistance at Q:

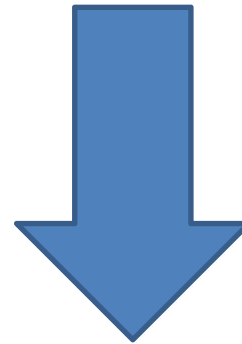
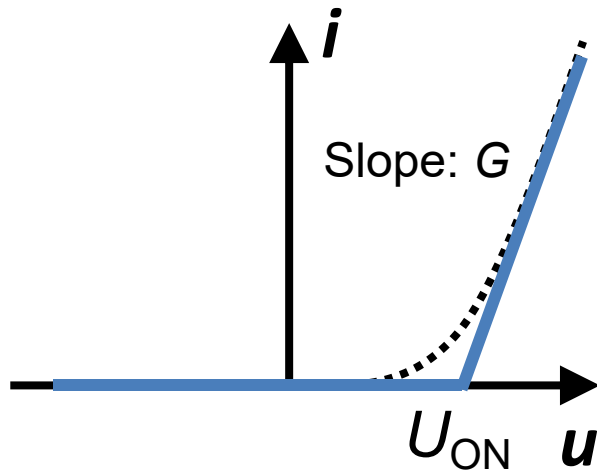
$$R_{DC} = \frac{U_Q}{I_Q}$$

AC/dynamic resistance at Q:

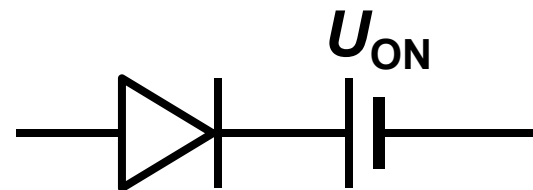
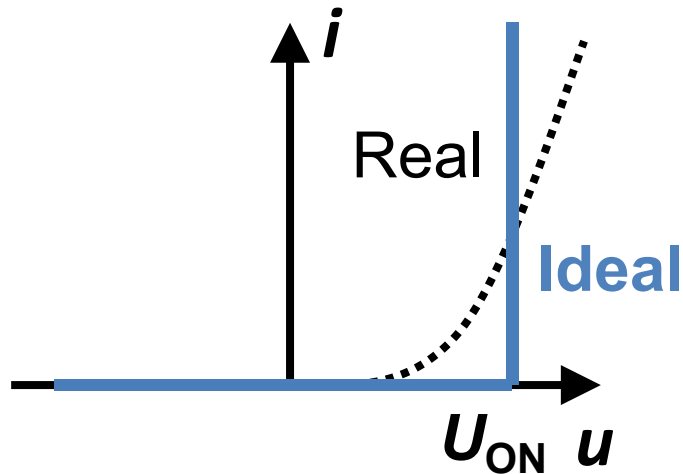
$$R_{AC} = \left. \frac{du}{di} \right|_{U_Q} = \frac{1}{G}$$

Offset-resistance model 折线模型

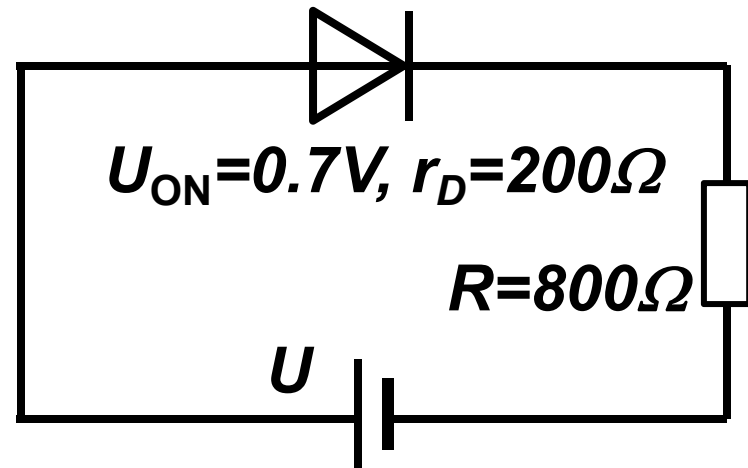




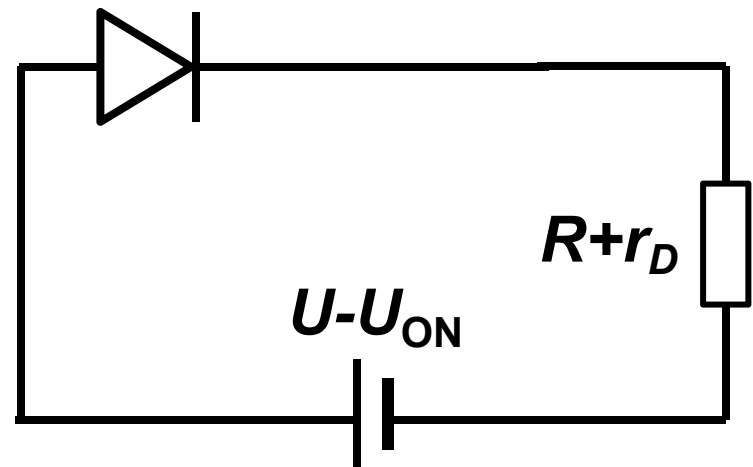
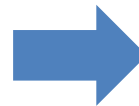
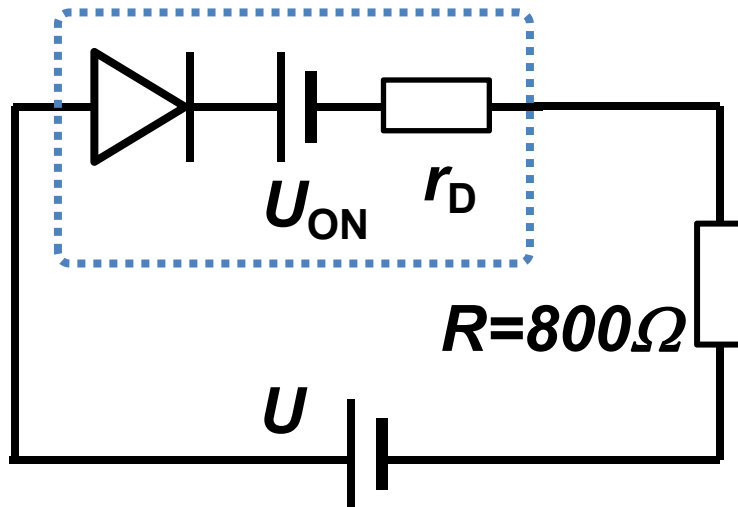
When $r_D = 0$:

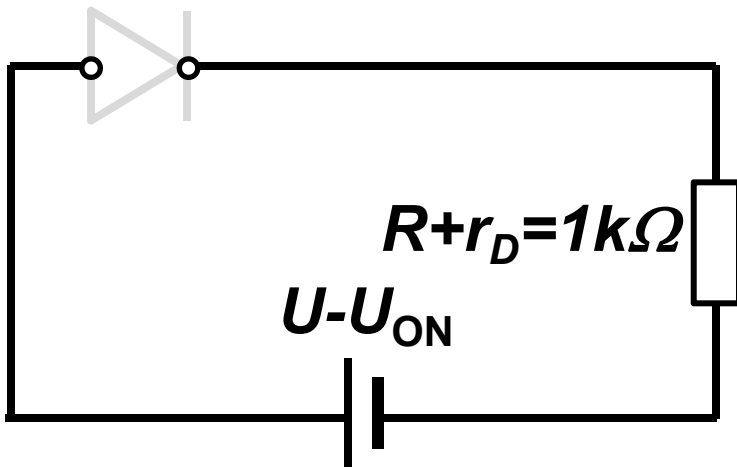
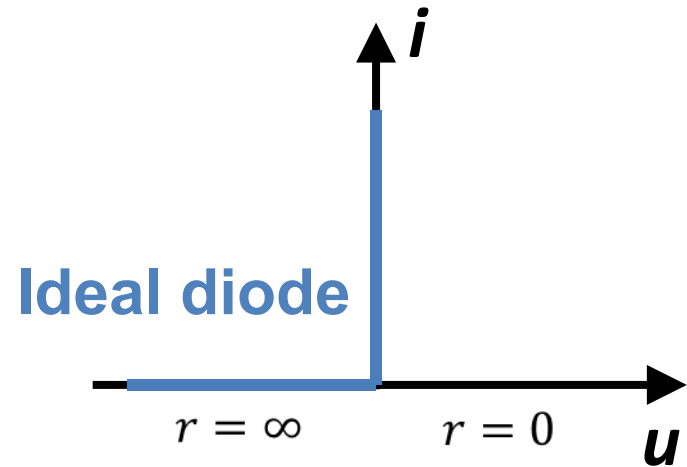
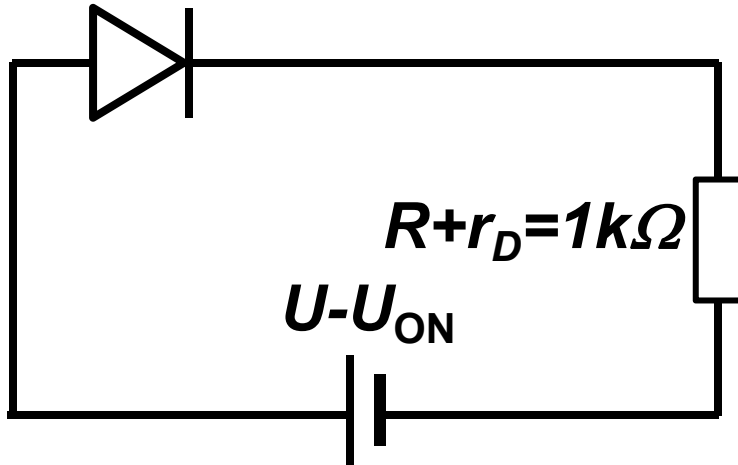


Example:



[Q1] Using the broken-line mode, ask: when $U=0.5V, 1.7V$, and $-1.3V$, the current in the circuit?



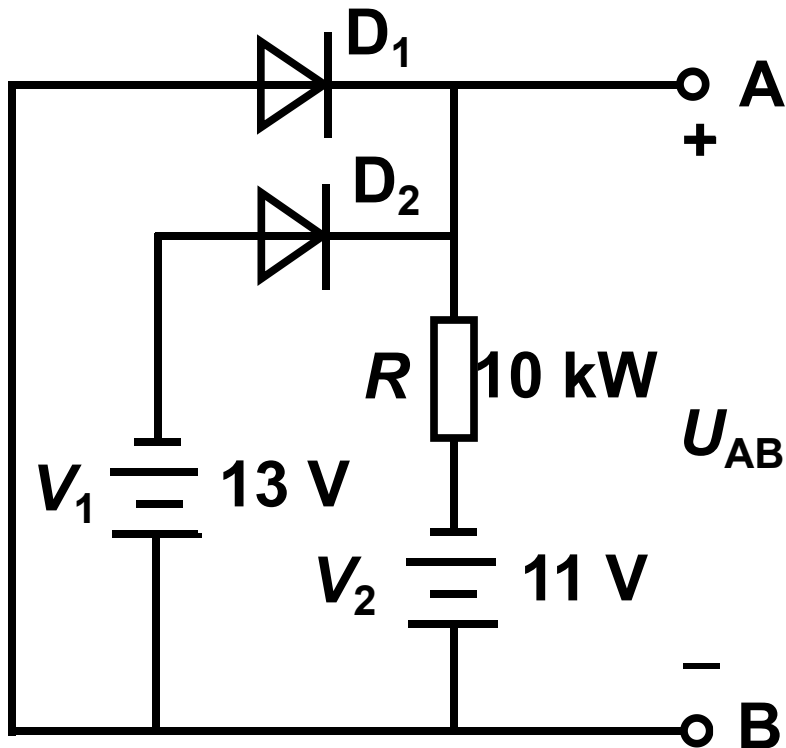


Assume diode is off

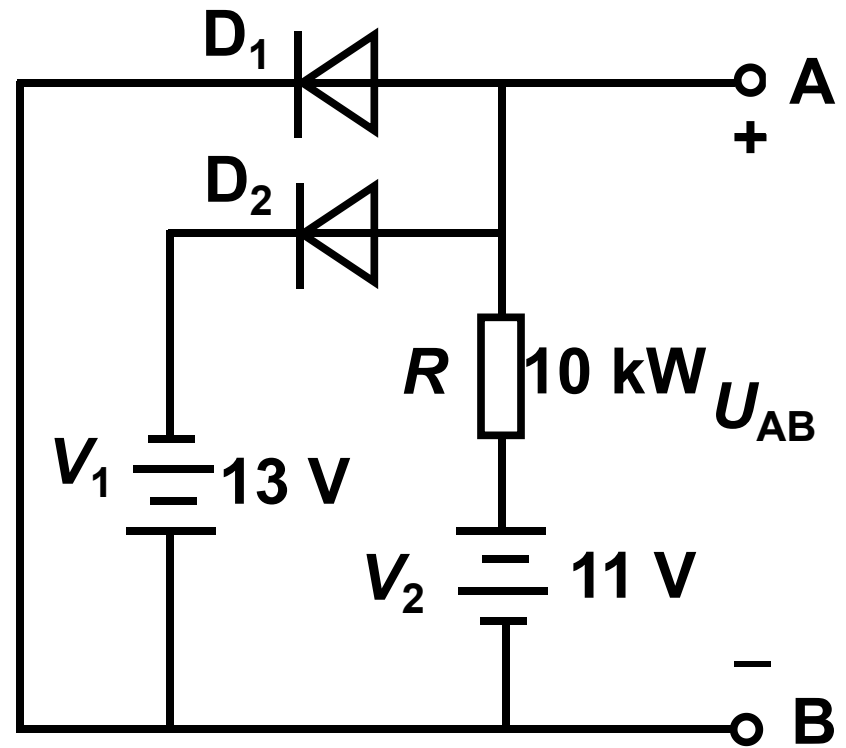
1. $U=-1.3V$, $U-U_{ON}=-2V$, $i=0$;
2. $U=0.5V$, $U-U_{ON}=-0.2V$, $i=0$;
3. $U=1.7V$, $U-U_{ON}=1V$,

Assumption is wrong, and diode is on: $i=(U-U_{ON})/(R+r_D)=1mA$;

[Q2] All the diode is ideal. Ask: (1) the diode D_1 and D_2 is on or off. (2) The value of U_{AB} .



(a)



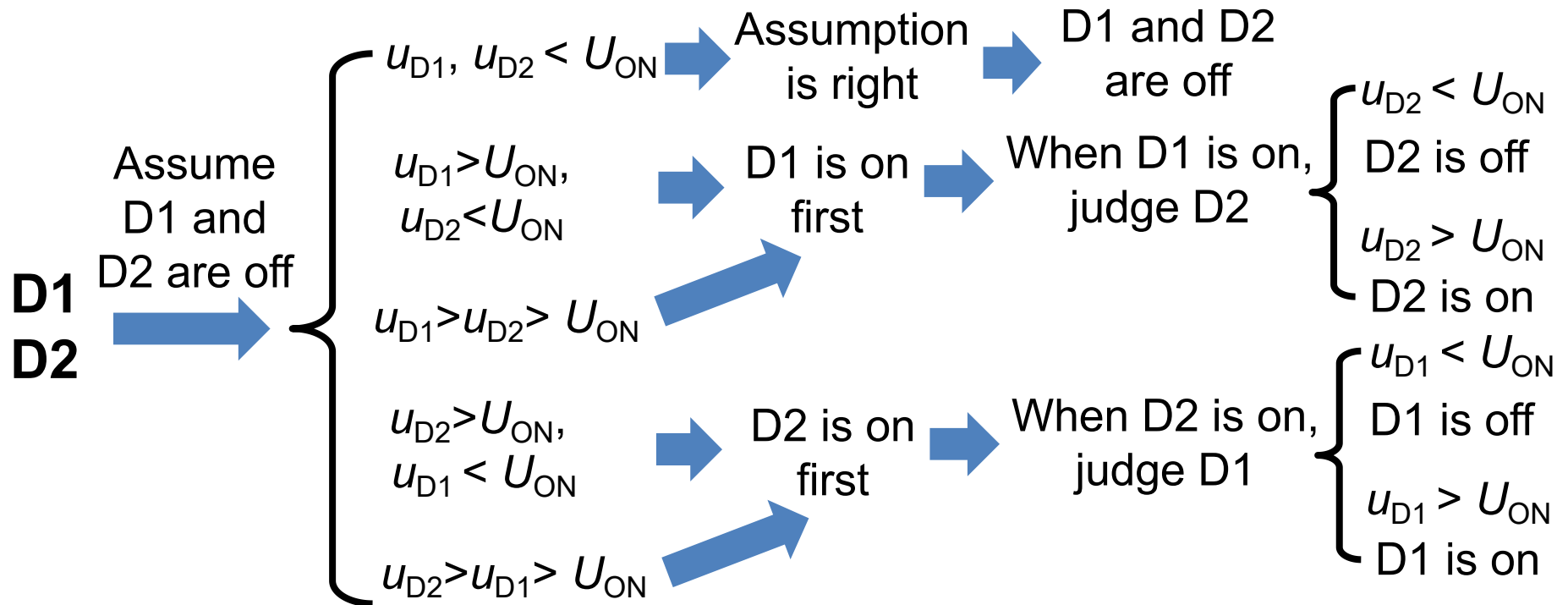
(b)

[Answer] How to judge diode on or off.

1. One diode: assume diode is off, and analyze the circuit. If the $u_{\text{Diode}} > U_{\text{ON}}$, diode is on. If the $u_{\text{Diode}} < U_{\text{ON}}$, diode is off.

2. If there are two identical diodes D_1 and D_2 in the circuit, assume both diodes are off, and analyze the circuit. Compare u_{Diode1} with u_{Diode2} . The diode with higher u_{Diode} is on first. With one diode on, then judge the other diode.

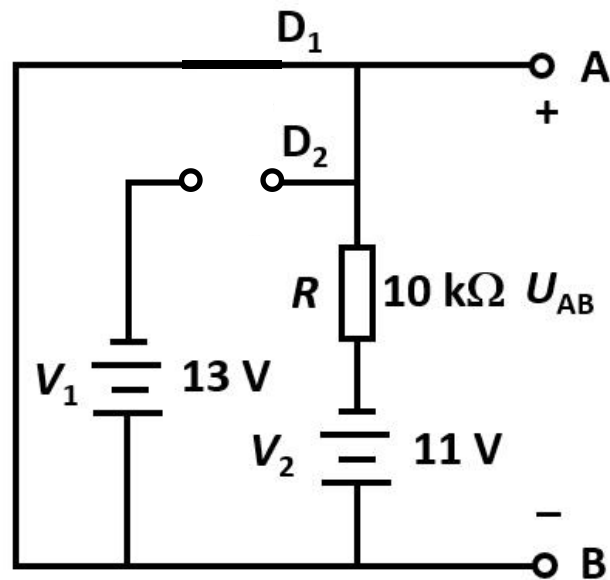
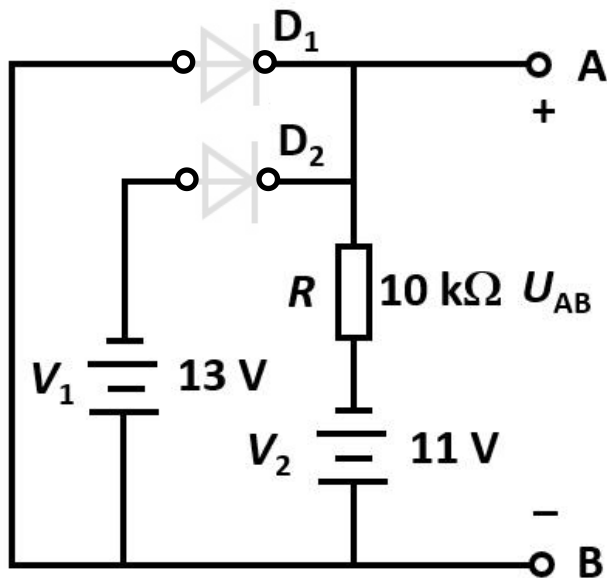
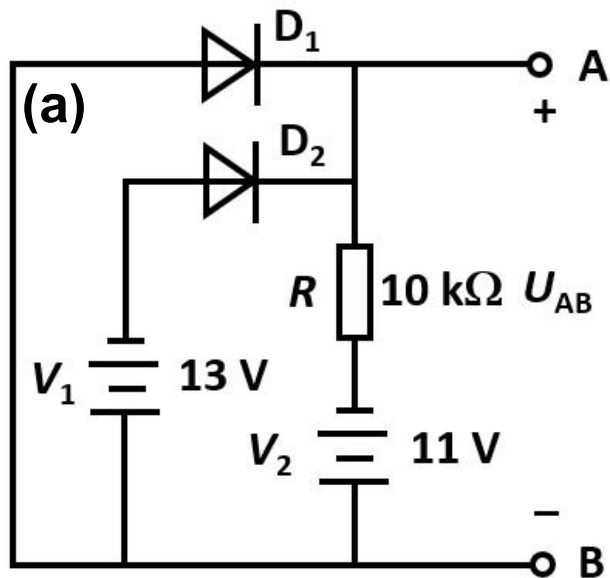
[Answer] How to judge diode on or off.



[Answer] 判断电路中二极管工作状态的方法

1. 断开二极管，分析电路断开点的开路电压。如果该电压能使二极管正偏，且大于二极管的开启电压，二极管导通；否则二极管截止。

2. 如果电路中有两个二极管，利用方法1分别判断各个二极管两端的开路电压，开路电压高的二极管优先导通；当此二极管导通后，再根据电路的约束条件，判断另一个二极管的工作状态。



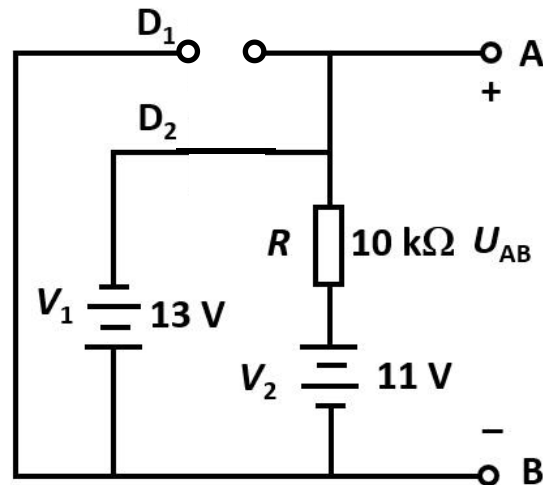
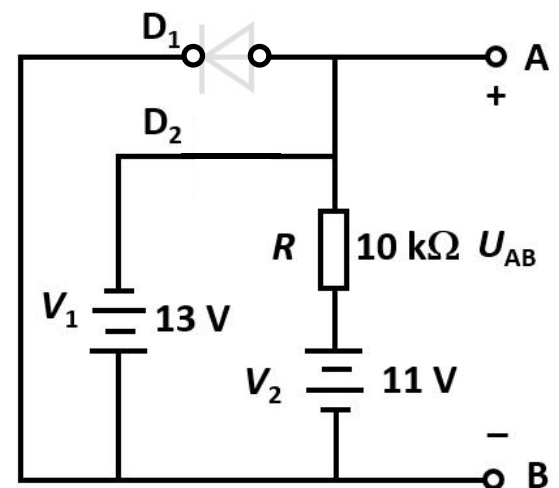
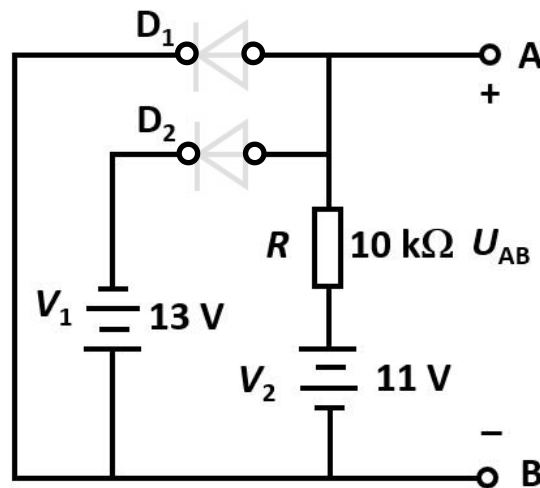
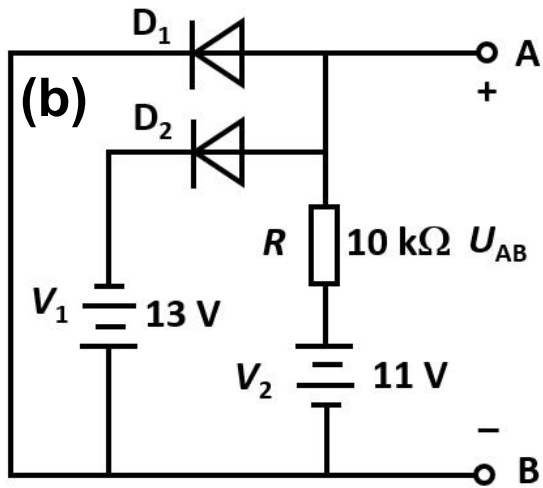
**Fig.(a): voltage of D_1 , D_2 is:
11 V, -2 V**

D_1 on, assume D_2 off

Voltage of D_2 is: -13 V

D_2 off

$U_{AB} = 0$ V



**Fig.(b): voltage of D_1 , D_2 is:
11 V, 24 V**

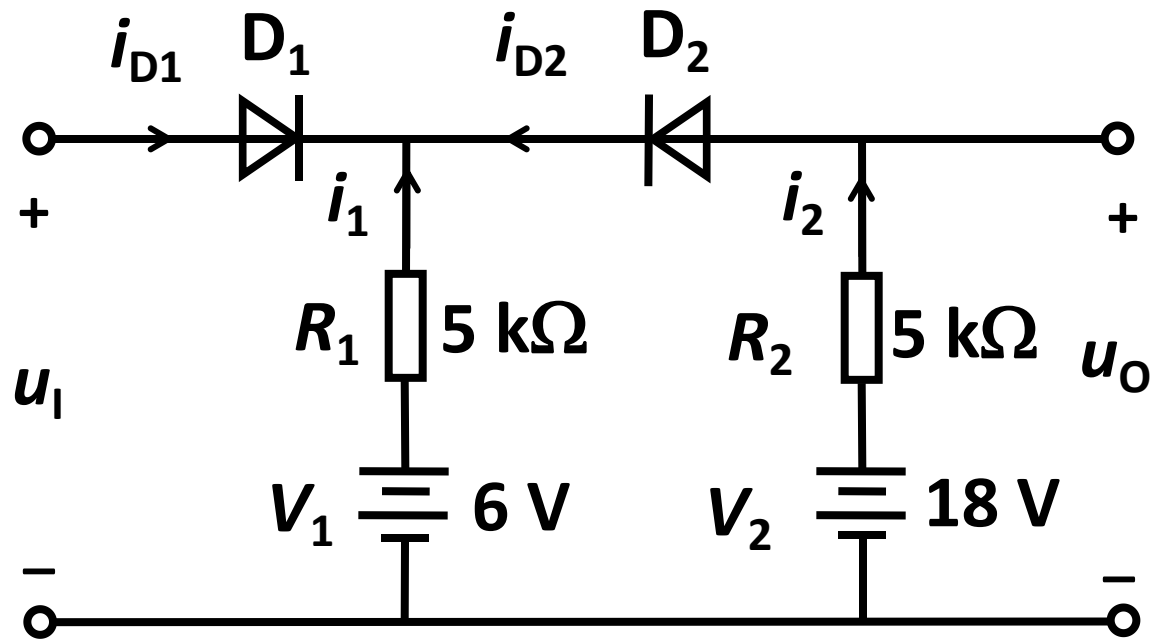
D_2 is on first

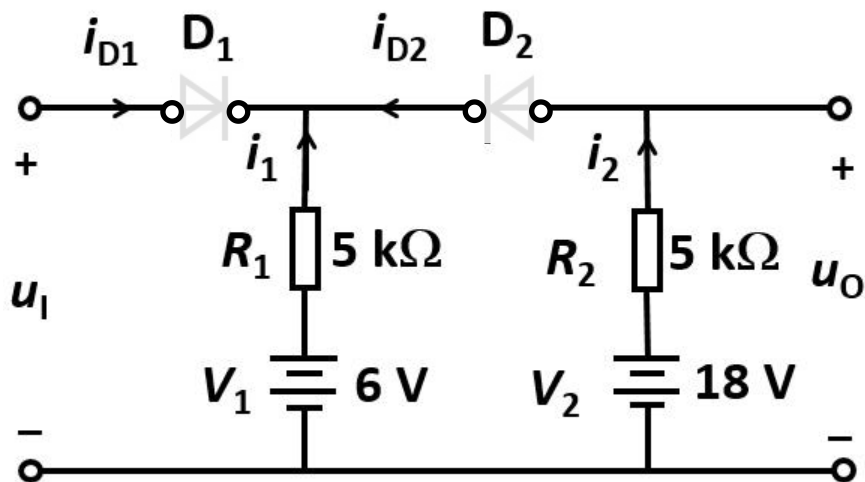
Assume D_1 is off

Voltage of D_1 is -13 V

D_1 off $U_{AB} = -13\text{ V}$

[Q3] D_1 and D_2 are ideal, the input voltage u_i change from 0V to 30 V. Please draw the voltage transmission curve ($u_o \sim u_i$) 电压传输特性曲线.





The first method

1.1 When $u_I \leq 12$ V, D_1 off, D_2 on:

$$i_2 = -i_1 = \frac{18 - 6}{5 + 5} \text{ mA} = 1.2 \text{ mA}$$

$$u_O = V_2 - i_2 R_2 = 12 \text{ V}$$

[Ans] Assume D_1 and D_2 are off:

$$V_{D1} = u_I - 6$$

1.2 When $12 \text{ V} < u_I < 18 \text{ V}$, D_1 , D_2 on

$$V_{D2} = 18 - 6 = 12 \text{ V}$$

$$u_O = u_I$$

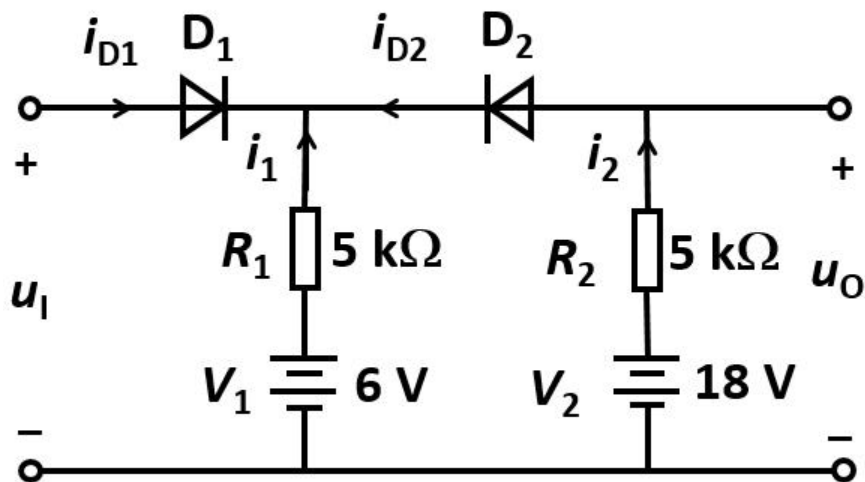
1. When $u_I < 18 \text{ V}$, D_2 is on first 2. When $18 \text{ V} \leq u_I$, D_1 is on first

When D_2 is on, voltage for D_1 : $V_{D2} = 18 - u_I < 0$, D_2 off

$$V_{D1} = u_I - 6 + 5i_1 = u_I - 12$$

$$i_2 = 0$$

$$u_O = V_2 = 18 \text{ V}$$



[Ans] Assume D_1 , D_2 are on:

$$i_{D2} = i_2 = \frac{V_2 - u_I}{R_2} = \frac{18 - u_I}{5}$$

$$i_{D1} = -i_1 - i_{D2}$$

$$= -\frac{V_1 - u_I}{R_1} - \frac{V_2 - u_I}{R_2} = \frac{2u_I - 24}{5}$$

When D_1 is on: $u_I > 12 \text{ V}$

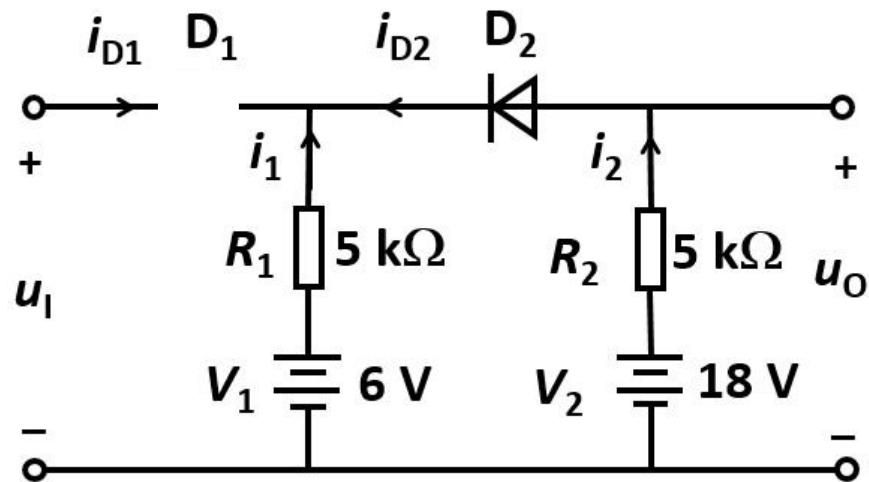
When D_2 is on: $u_I < 18 \text{ V}$

The second method

1. When $12 \text{ V} < u_I \leq 18 \text{ V}$, D_1 on, D_2 on

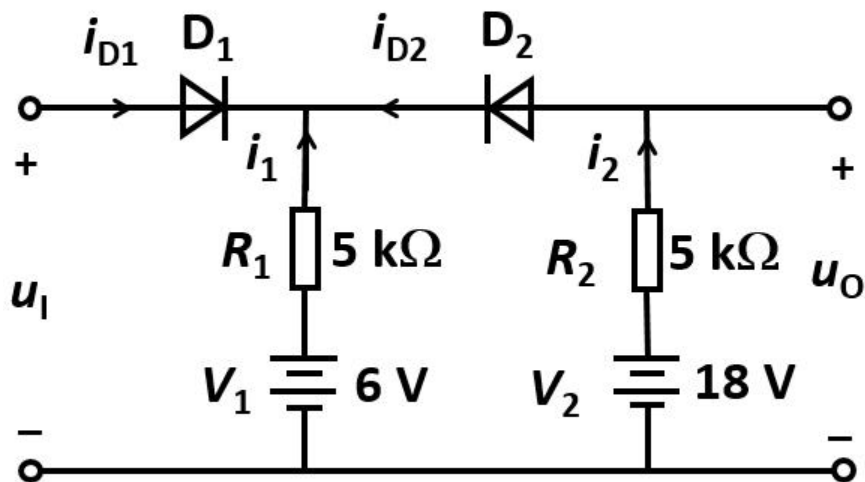
$$u_O = u_I$$

2. When $u_I \leq 12 \text{ V}$, D_1



$$i_2 = -i_1 = \frac{18 - 6}{5 + 5} \text{ mA} = 1.2 \text{ mA}$$

D_2 on $u_O = V_2 - i_2 R_2 = 12 \text{ V}$



[Ans] Assume D_1 , D_2 are on:

$$i_{D2} = i_2 = \frac{V_2 - u_I}{R_2} = \frac{18 - u_I}{5}$$

$$i_{D1} = -i_1 - i_{D2}$$

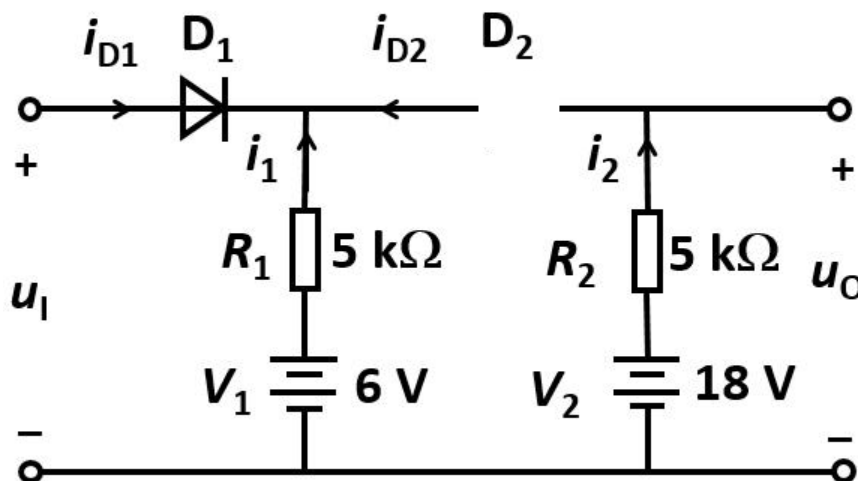
$$= -\frac{V_1 - u_I}{R_1} - \frac{V_2 - u_I}{R_2} = \frac{2u_I - 24}{5}$$

When D_1 is on: $u_I > 12 \text{ V}$

When D_2 is on: $u_I < 18 \text{ V}$

The second method

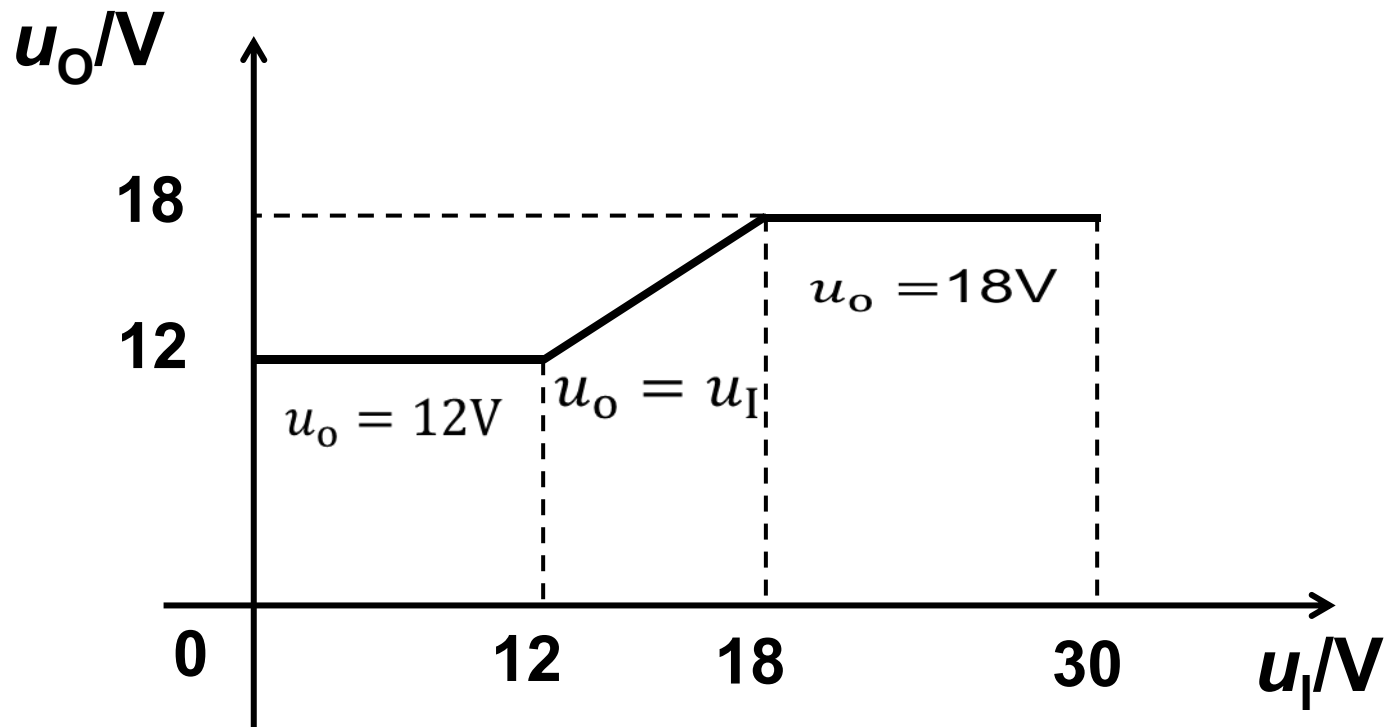
3. When $18 \text{ V} < u_I$, D_2 off



$$i_2 = 0$$

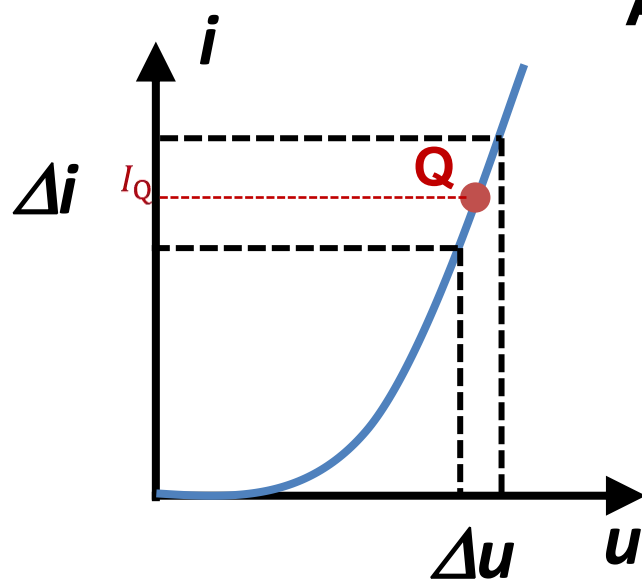
$$u_O = V_2 = 18 \text{ V}$$

$u_o \sim u_i$ curve



Small signal model 小信号模型、微变模型

This model is only for AC signal 交流信号



AC/dynamic resistance at Q point:

$$\frac{1}{r_D} = \frac{\Delta i}{\Delta u} \approx \frac{di}{du}$$

$$= \frac{d[I_s(e^{\frac{u}{U_T}} - 1)]}{du} = \frac{dI_s e^{\frac{u}{U_T}}}{du} - \frac{dI_s}{du}$$

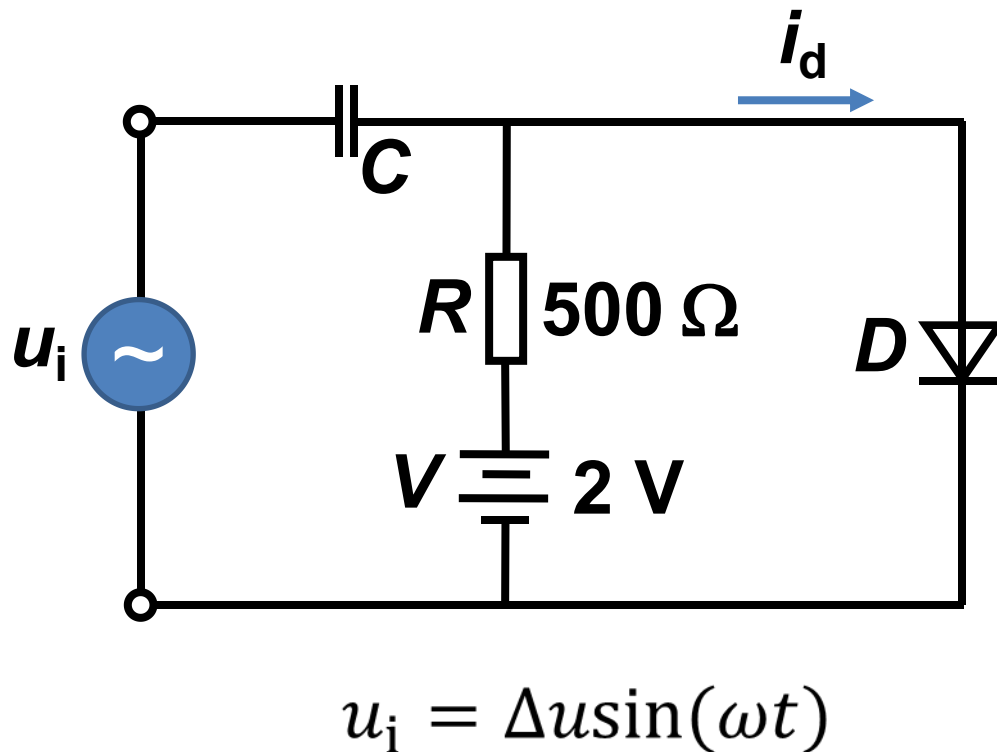
$$\approx \frac{I_s}{U_T} \cdot e^{\frac{u}{U_T}} \approx \frac{I_Q}{U_T}$$

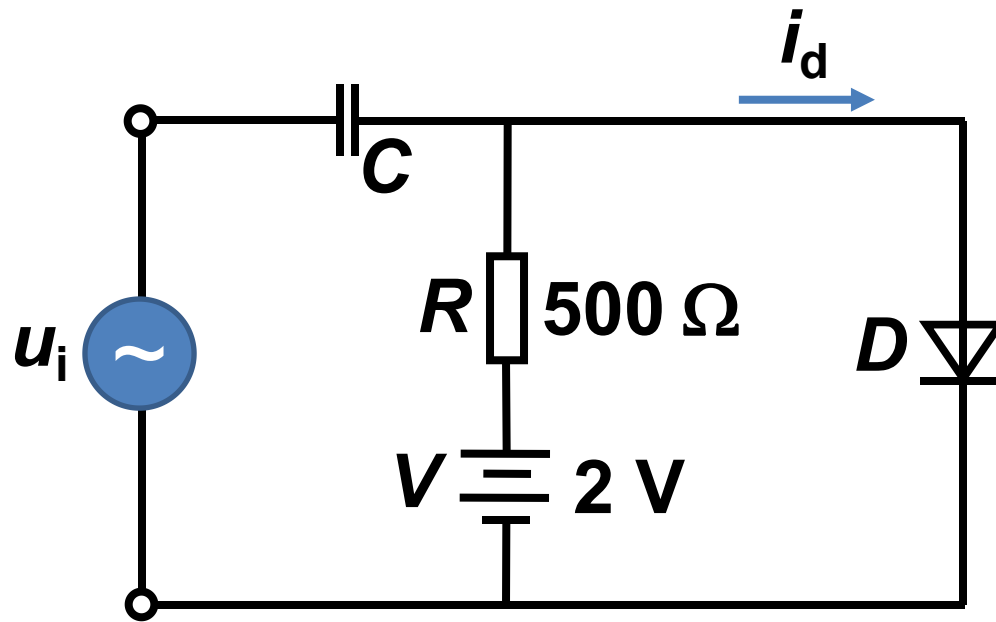
$$r_D = \frac{U_T}{I_Q}$$

Thermal voltage 热电压

$$U_T = \frac{KT}{q}$$

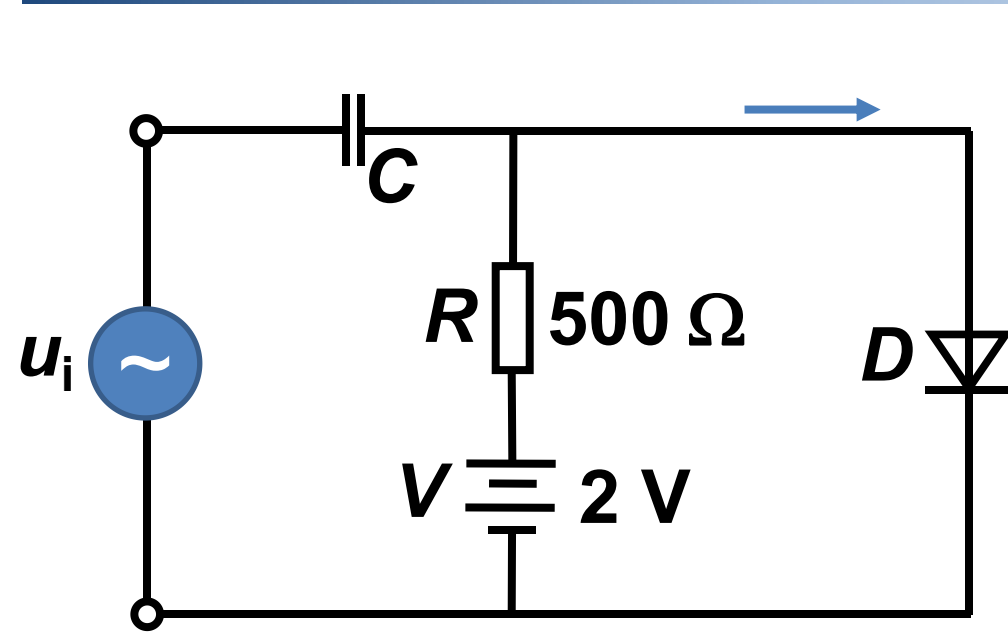
[Q4]: The working voltage of diode is 0.7 V; u_i is AC signal with amplitude of 10 mV; The capacitor has no resistance to AC signal; The thermal voltage $U_T=26$ mV. Ask: the amplitude of AC current i_d ?



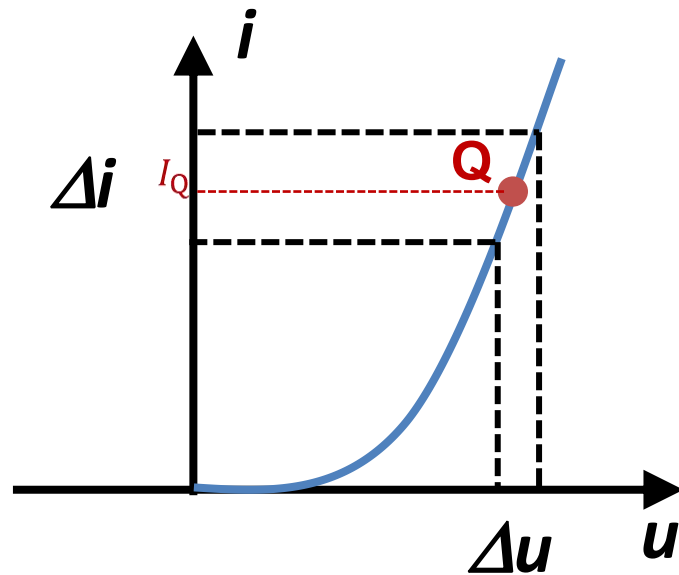
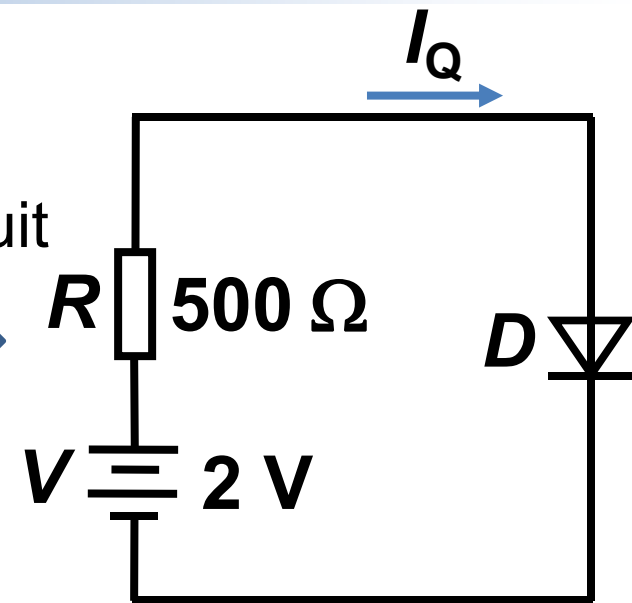


How to solve this problem?

Separate DC and AC circuit!

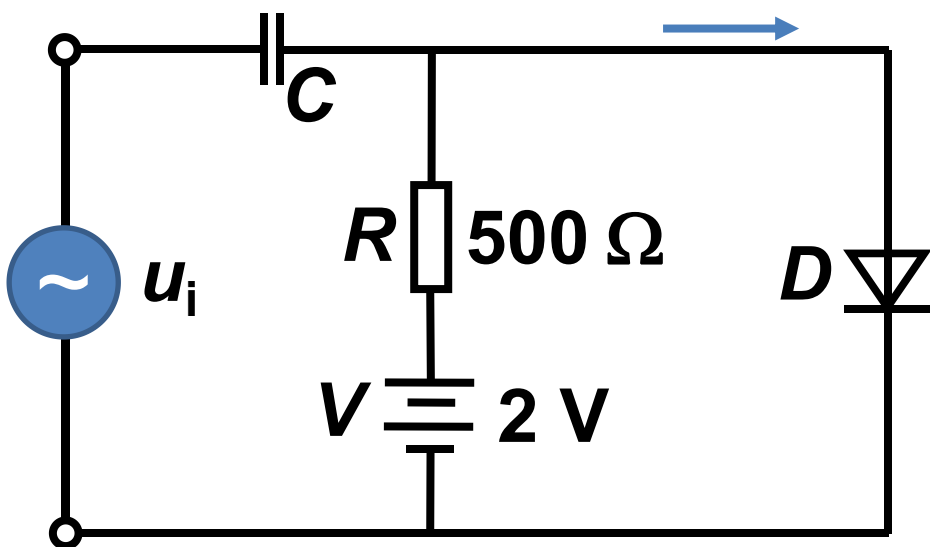


DC circuit

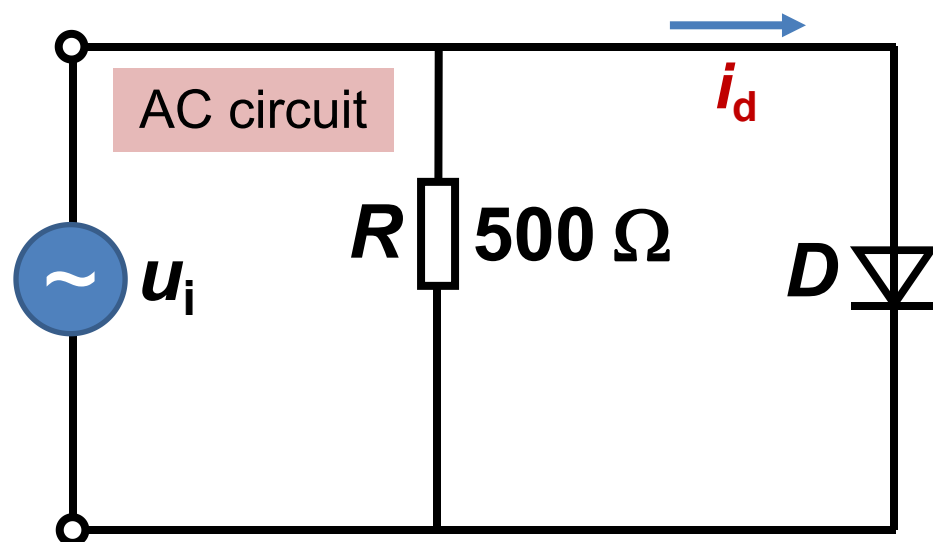
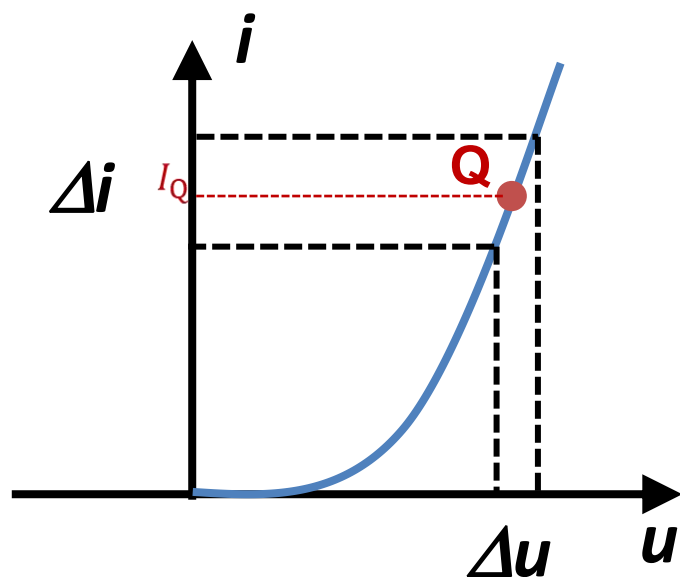


DC current:

$$\begin{aligned}
 I_Q &= \frac{V - U_{\text{work}}}{R} \\
 &= \frac{2 - 0.7}{500} \\
 &= 2.6\text{ mA}
 \end{aligned}$$



$$u_i = \Delta u \sin(\omega t)$$



AC resistance of diode:

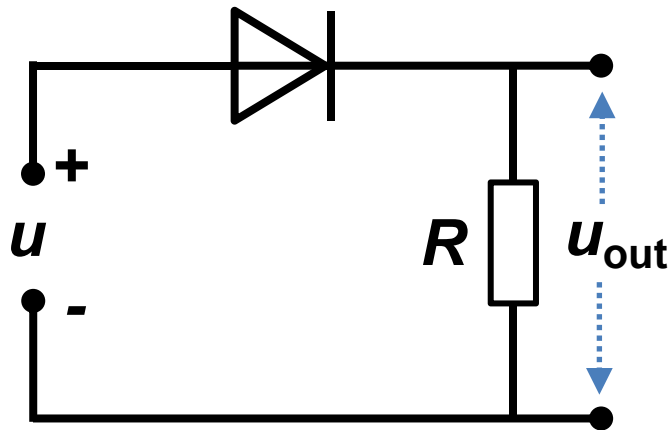
$$r_D = \frac{U_T}{I_Q} = \frac{26}{2.6} = 10 \Omega$$

Amplitude of AC current:

$$\Delta i = \frac{\Delta u}{r_D} = \frac{10 \text{ mV}}{10 \Omega} = 1 \text{ mA}$$

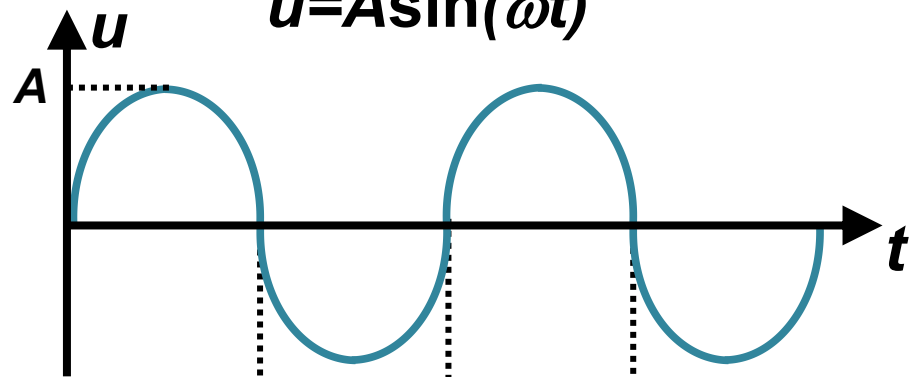
Application of diodes as rectifier

二极管在整流电路的应用

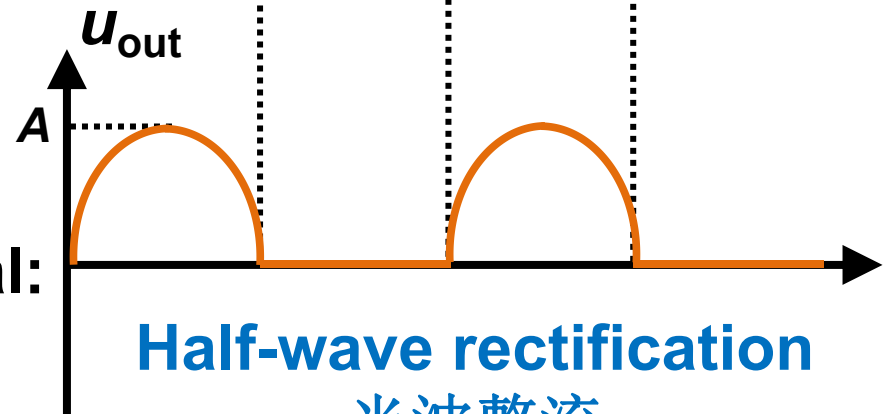


Input AC voltage signal:

$$u = A \sin(\omega t)$$



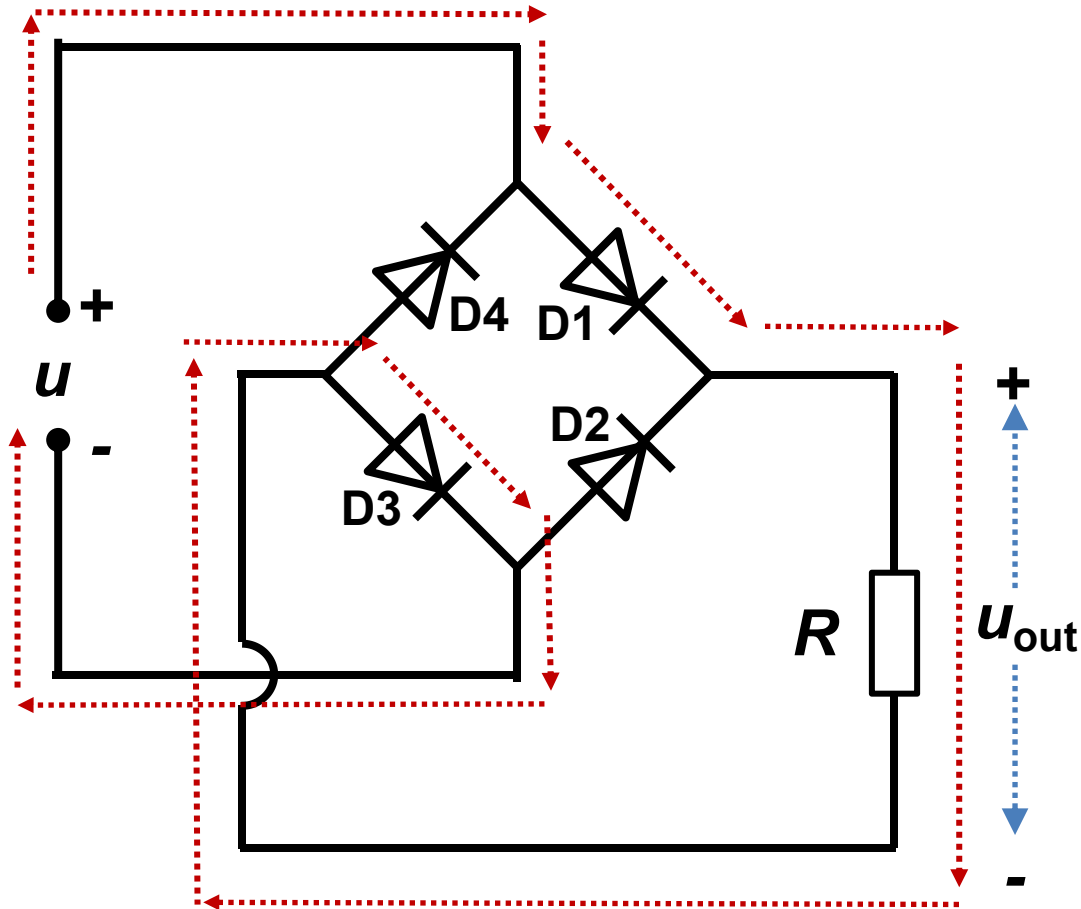
Output AC voltage signal:



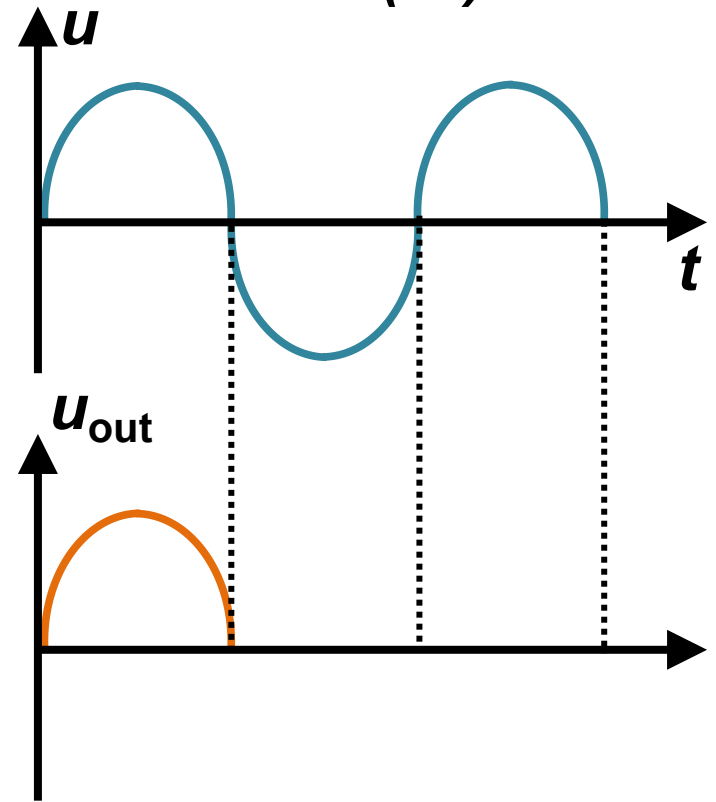
Half-wave rectification

半波整流

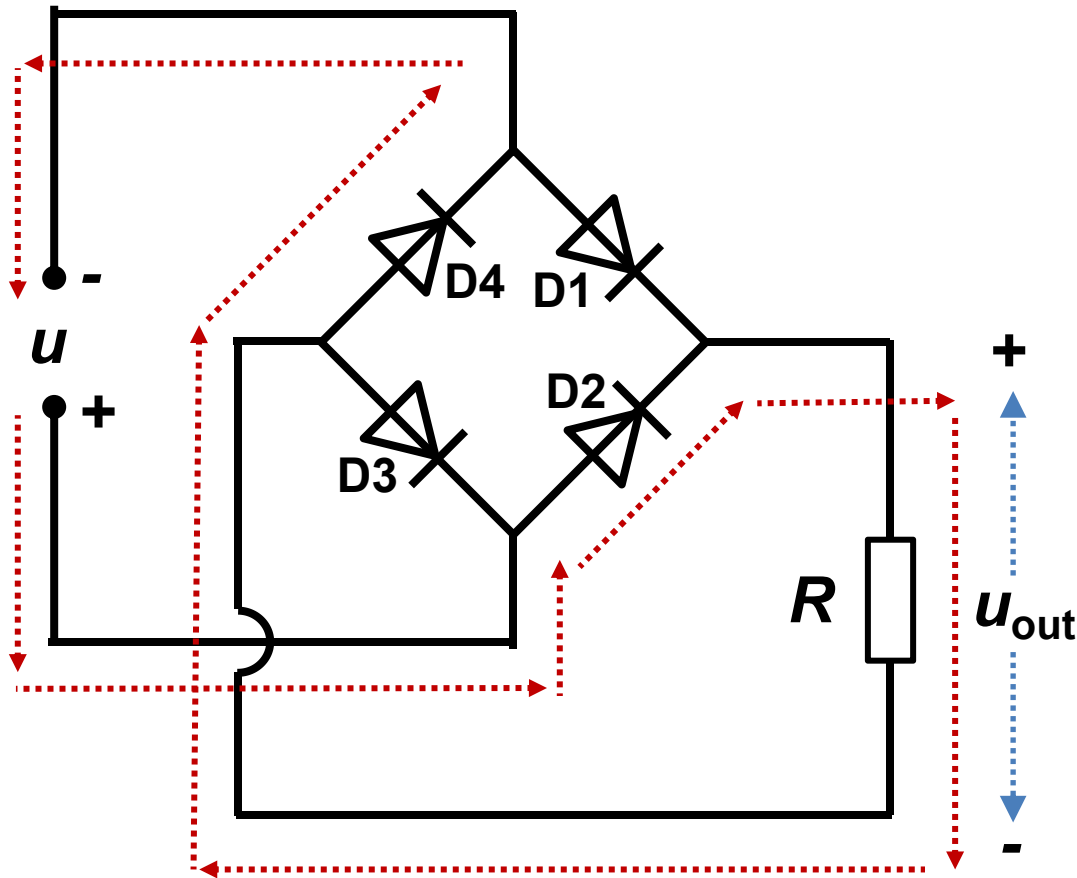
Full-wave rectification 全波整流



Input AC voltage signal:
 $u = A \sin(\omega t)$

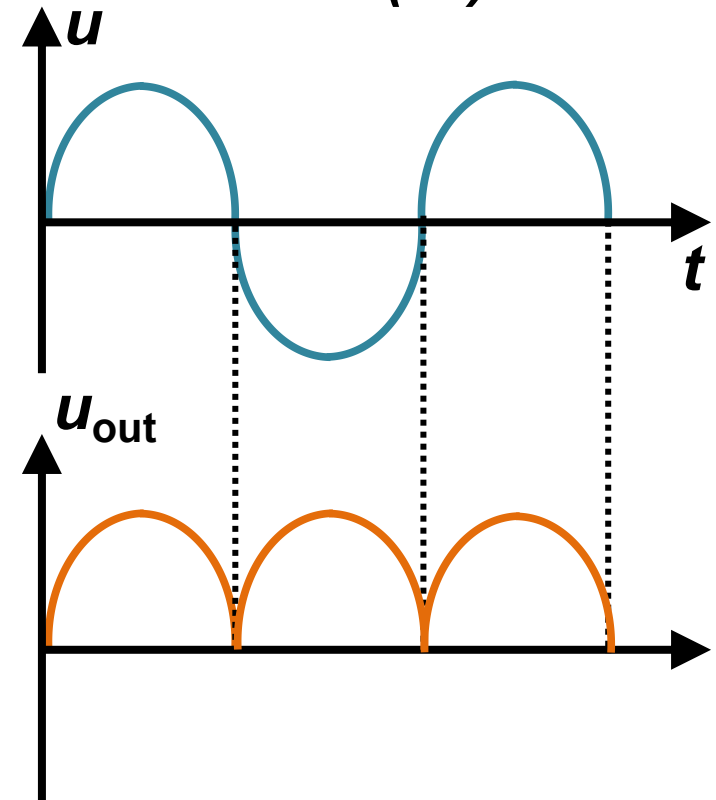


Full-wave rectification全波整流

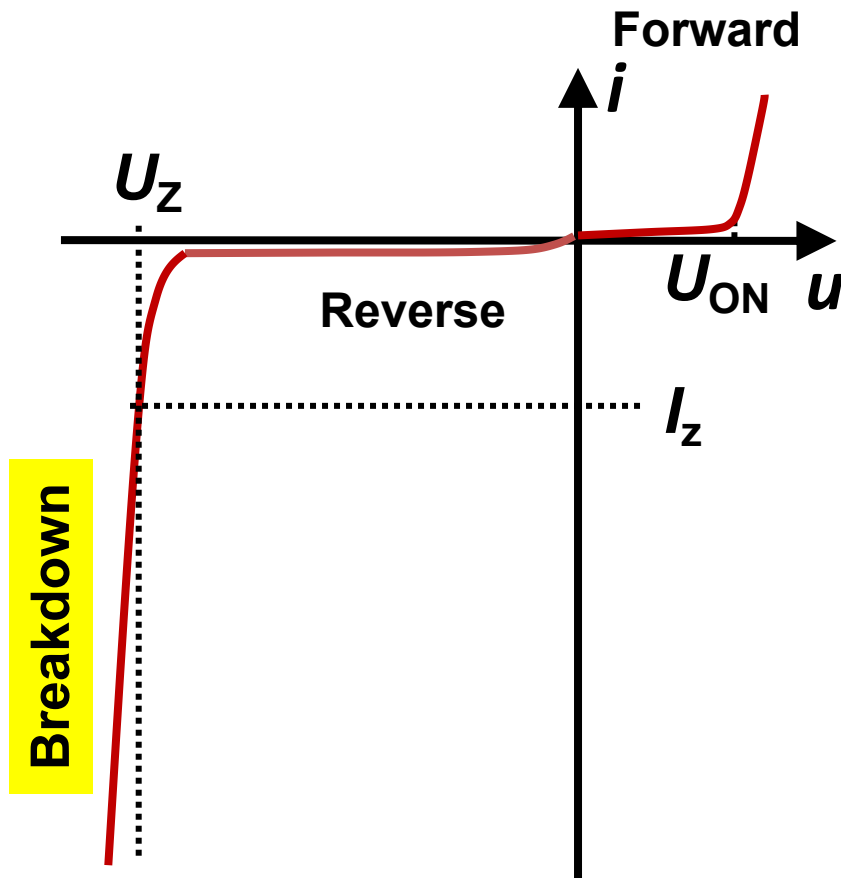


Input AC voltage signal:

$$u = A \sin(\omega t)$$



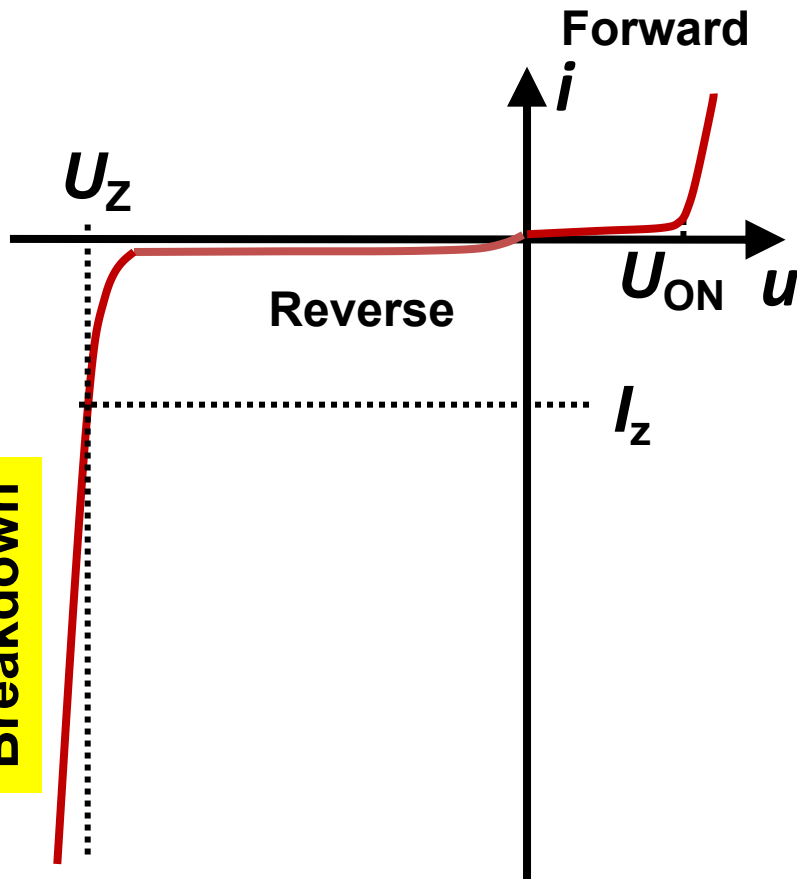
1.2.5 Voltage regulator diode 稳压二极管



Working principle:

I-V curve in **breakdown** region is very steep, which is **almost parallel** to y-axis.

The reverse current change significantly, while the voltage is almost a constant. Hence it can **stabilize the voltage**.



Reverse breakdown voltage $U_Z \gg$
Forward cut-in voltage U_{ON}

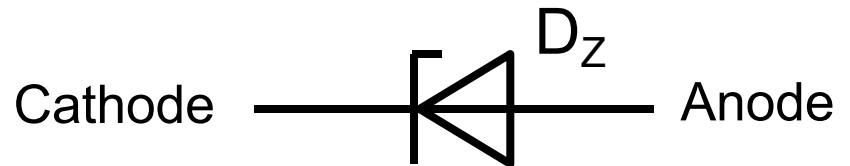
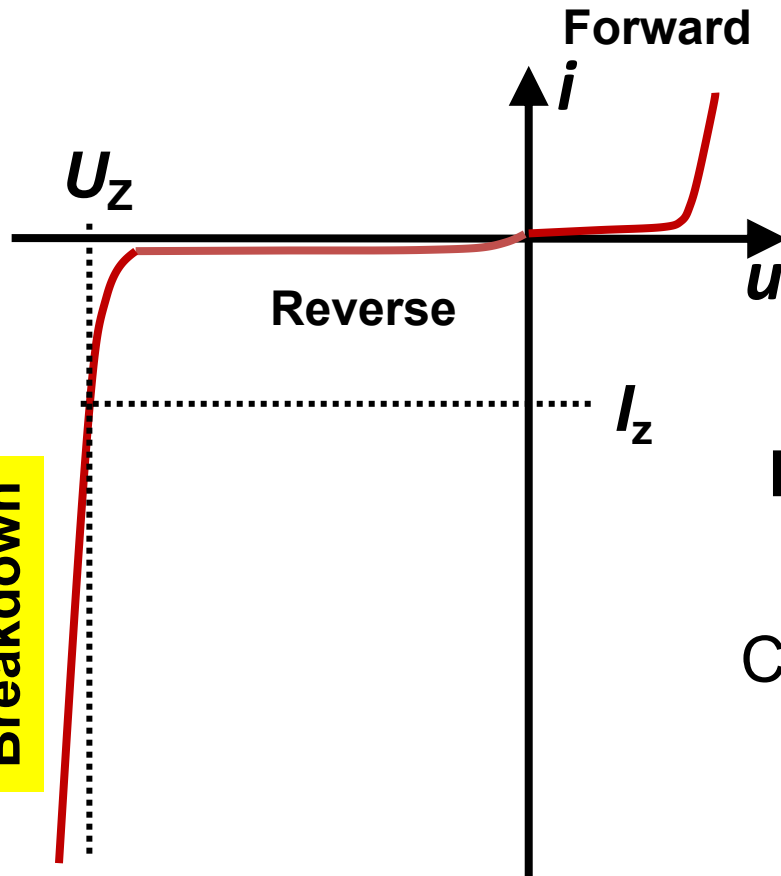
Take Si as an example:

$$U_Z > 4 \text{ V}$$

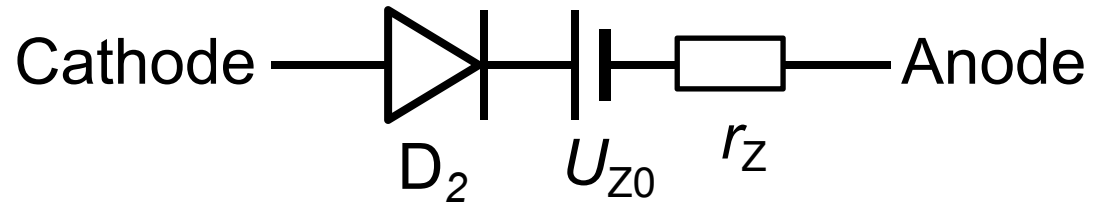
$$U_{ON} \sim 0.7 \text{ V}$$

Breakdown

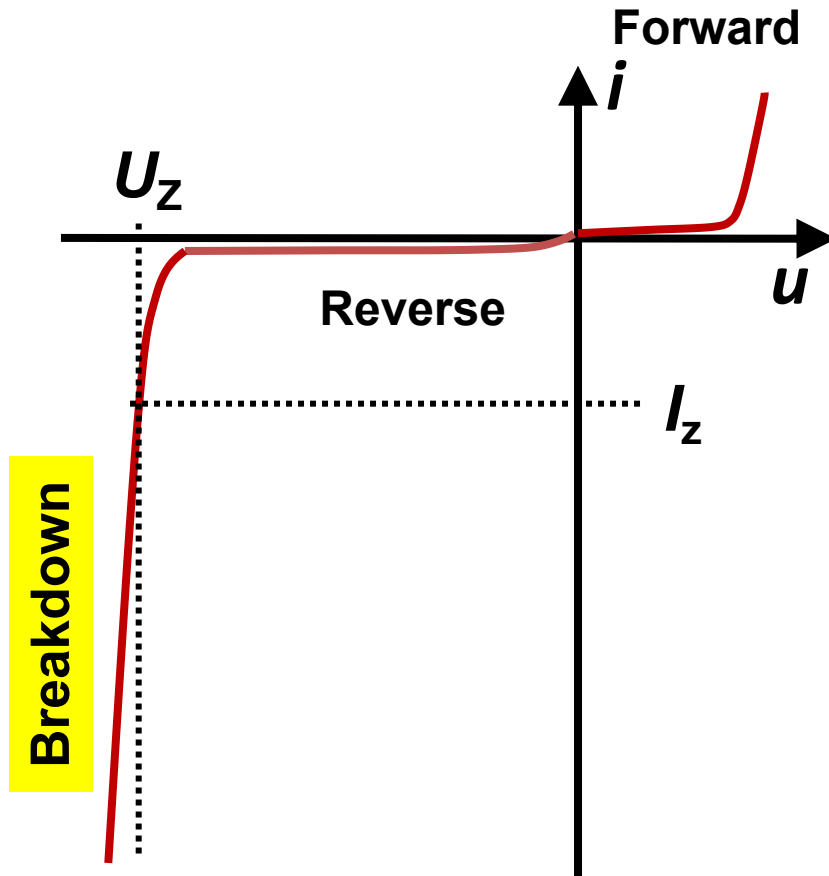
Symbol of voltage regulator diode:



Equivalent circuit: **Reverse biased**



For ideal regulator diode: $r_z=0$



Key parameters:

1. **Working current I_Z :** $I_{Zmax} > I_Z > I_{Zmin}$.

$I_Z > I_{Zmax}$: device breakdown permanently.

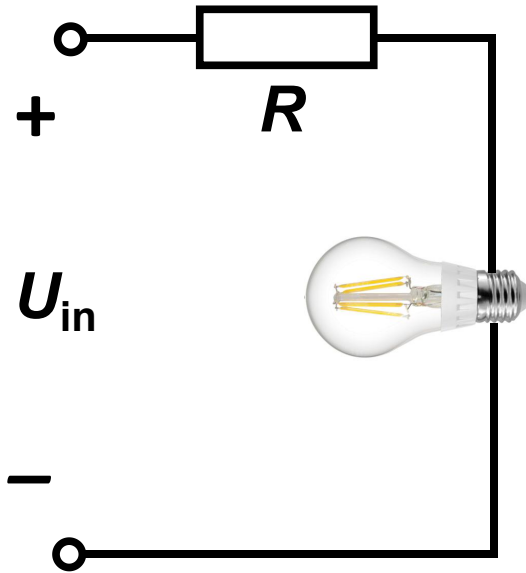
$I_Z < I_{Zmin}$, the voltage stabilization effect becomes worse.

2. **Working voltage U_Z :** almost a constant

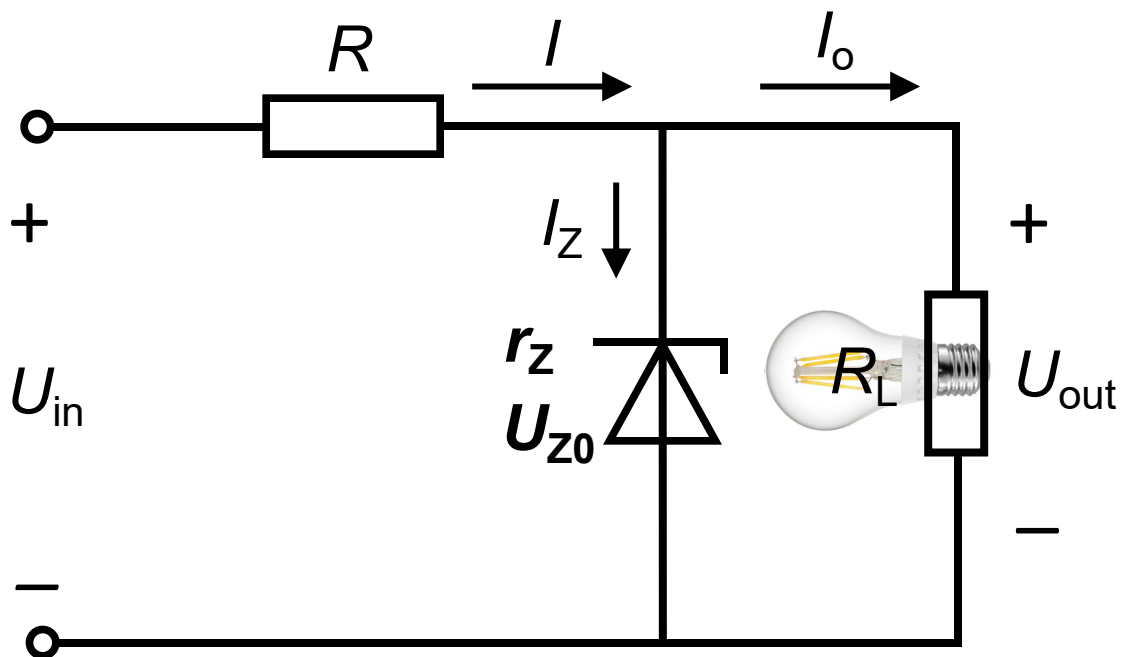
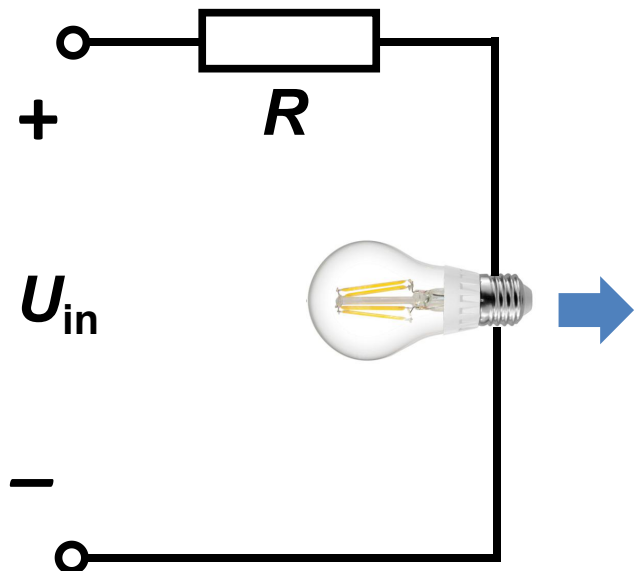
3. **Dynamic resistance r_Z :**
Smaller r_Z , better performance

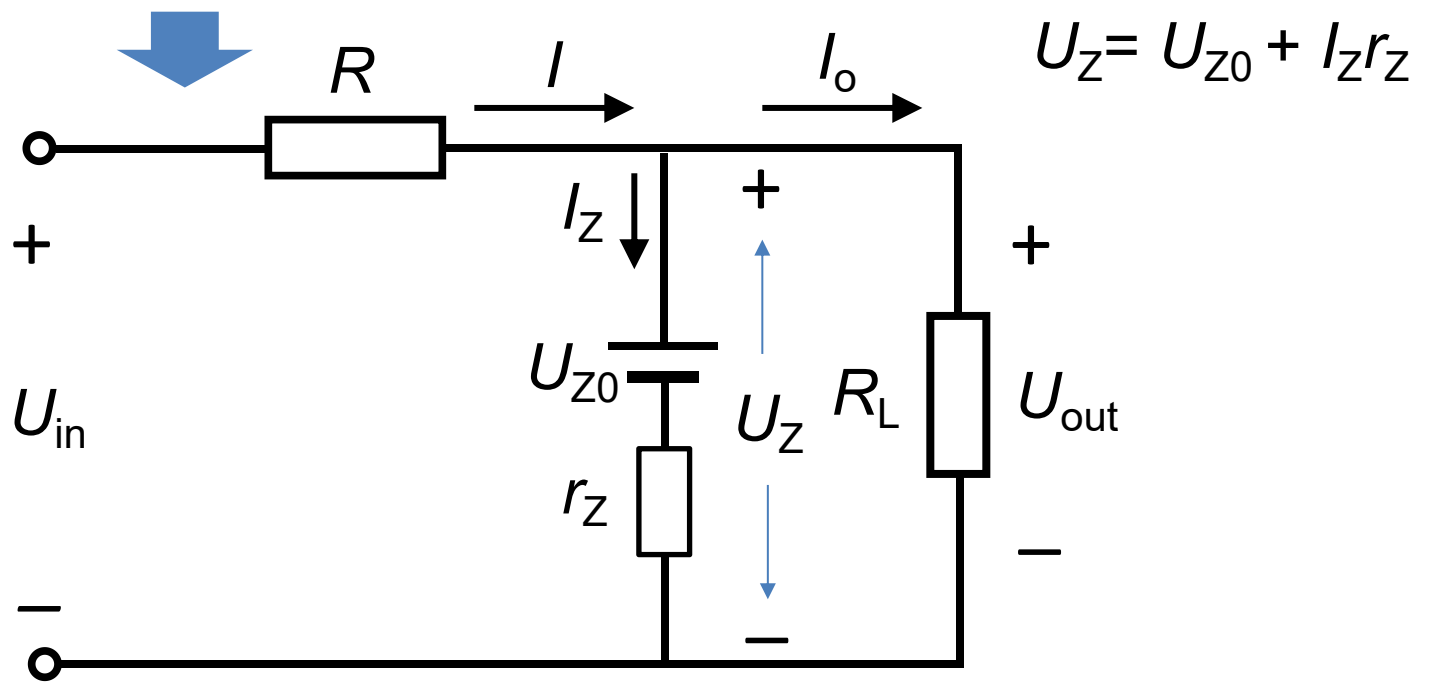
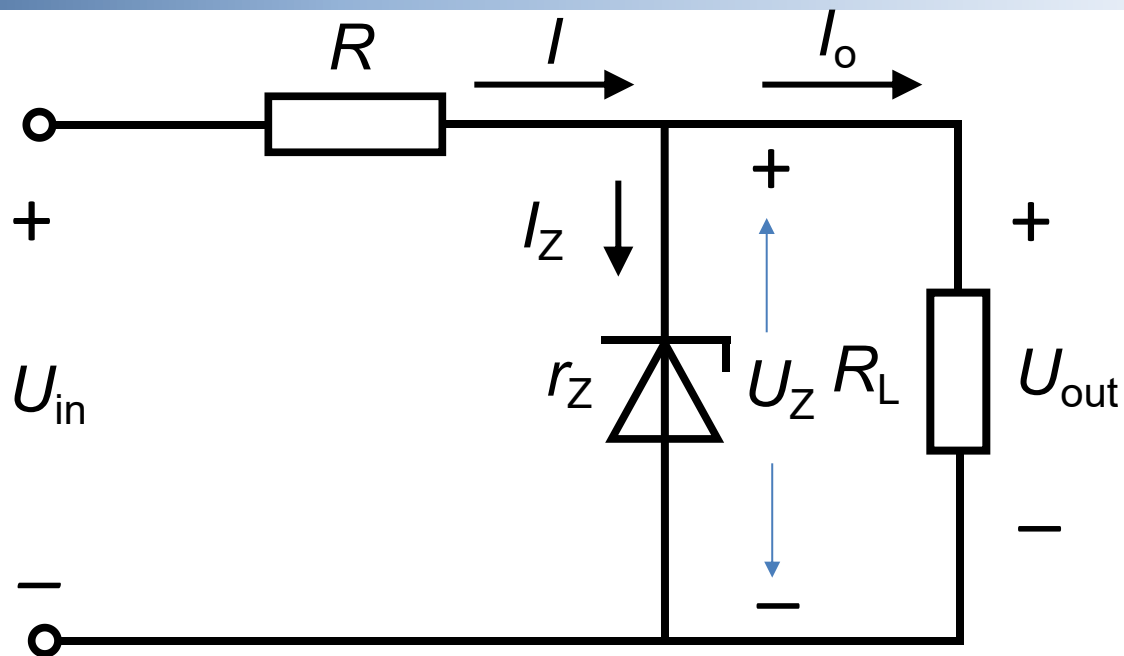


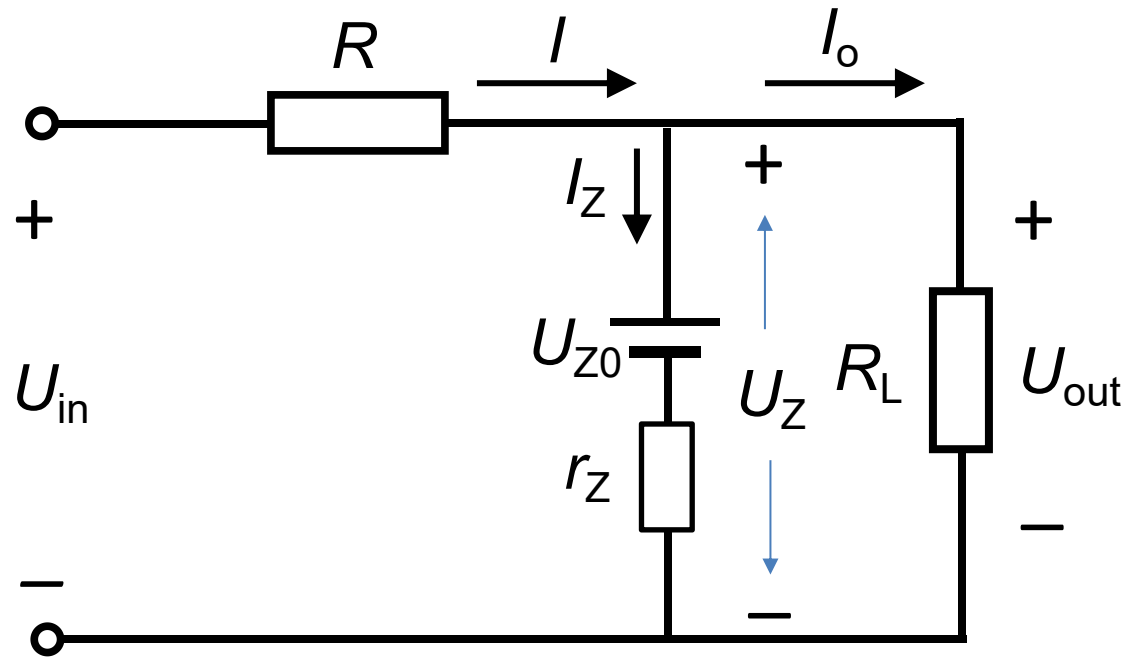
Working principle



Q: Input voltage U_{in} is unstable, and the bulb is flickering. How to resolve the problem?





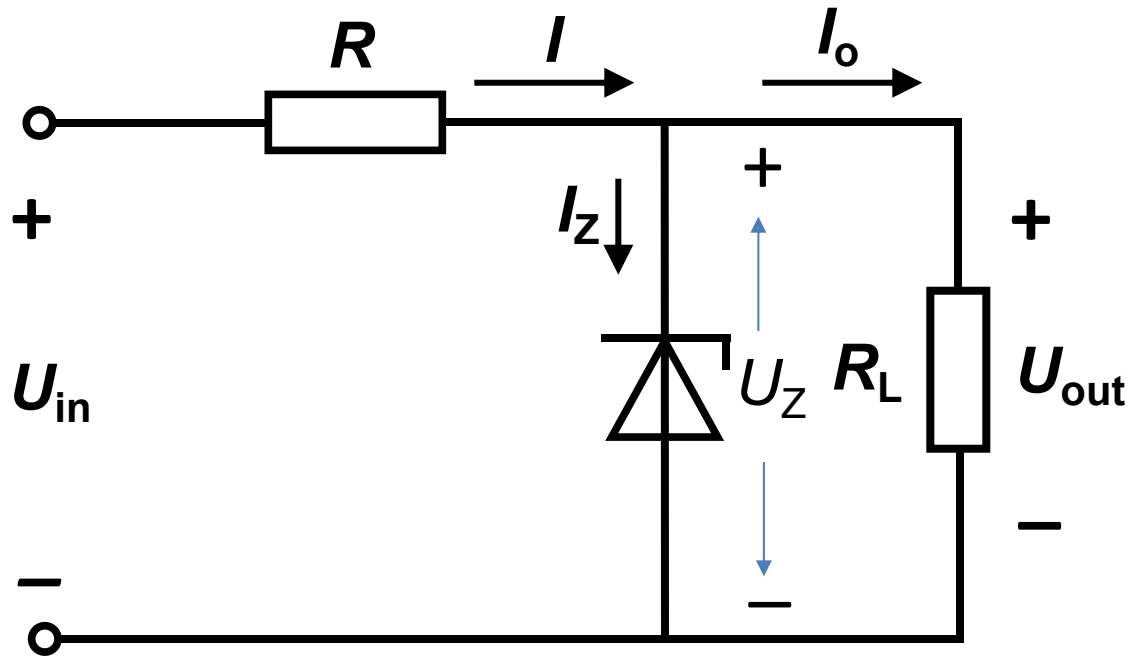


For ideal regulator diode, $r_Z=0$



$$U_{out} = U_{Z0}$$

R_L
unstable

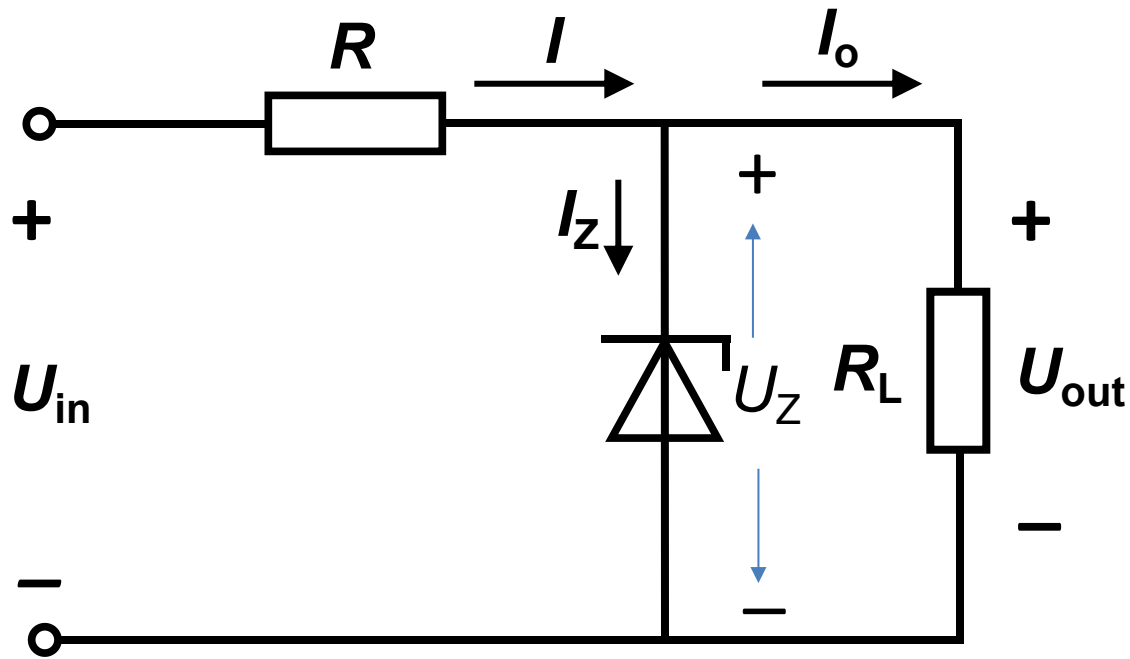


$R_L \uparrow \Rightarrow U_{out} \uparrow \Rightarrow U_Z \uparrow \Rightarrow I_Z \uparrow \uparrow \uparrow \Rightarrow I \uparrow \uparrow \uparrow$



Negative feedback! $U_{out} = U_{in} - IR \downarrow \leftarrow IR \uparrow \uparrow \uparrow$

U_{in}
unstable

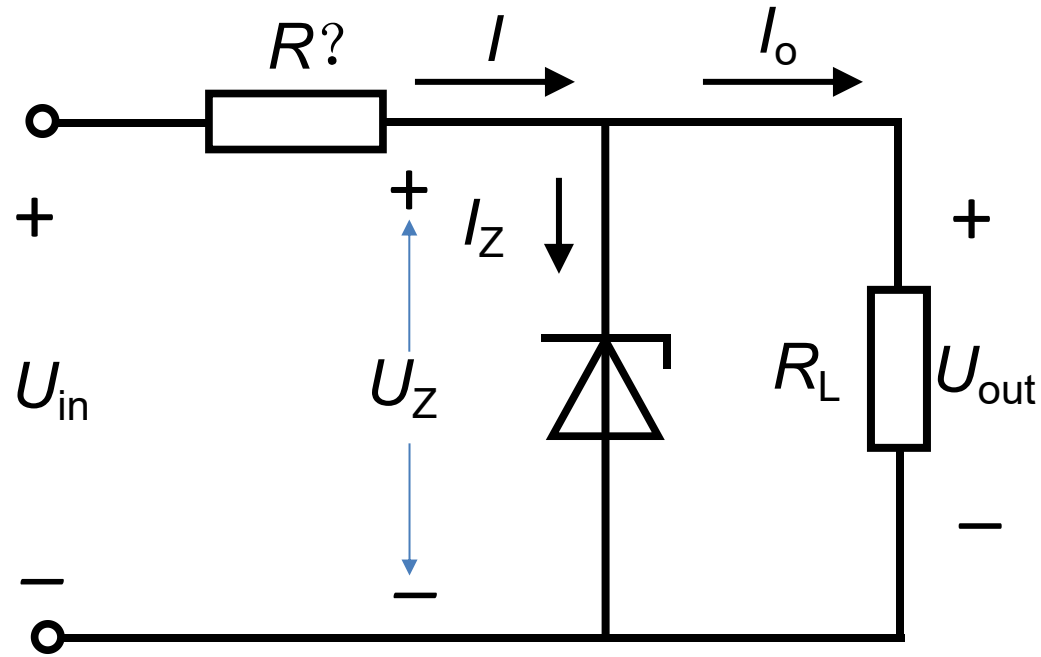


$U_{in} \uparrow \rightarrow U_{out} \uparrow \rightarrow U_Z \uparrow \rightarrow I_Z \uparrow \uparrow \uparrow \rightarrow I \uparrow \uparrow \uparrow$

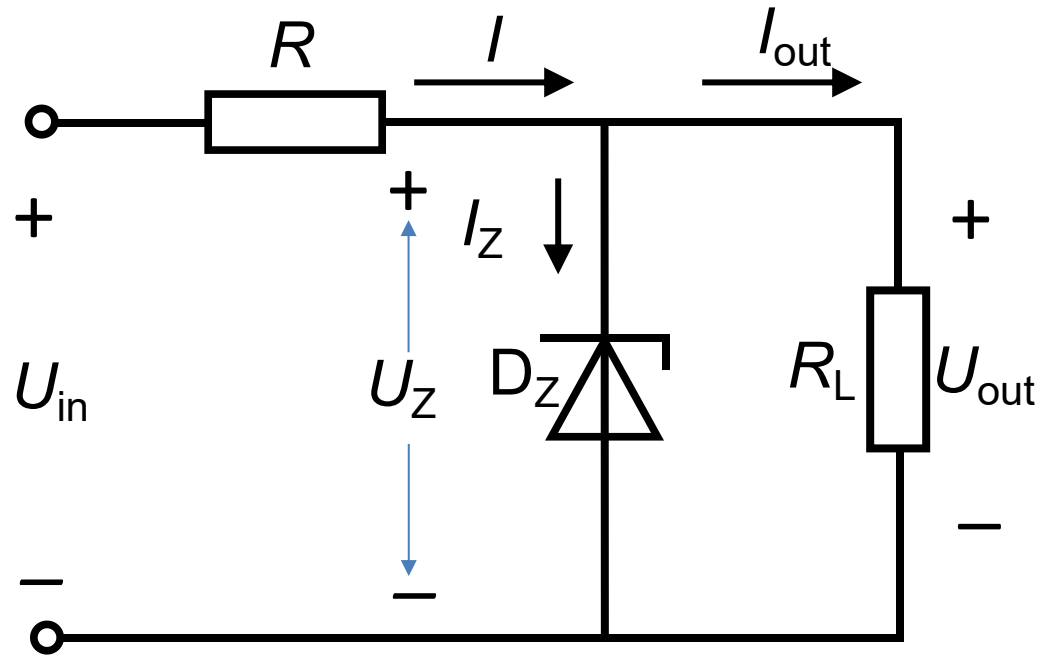


Negative feedback!

$U_{out} = U_{in} - IR \downarrow \leftarrow IR \uparrow \uparrow \uparrow$



[Q5] The working voltage of an ideal regulator diode is U_Z , the maximum and minimum working current is I_{ZMAX} and I_{ZMIN} . Ask: the value of resistance R ?



Assume voltage regulator is on

Output voltage: $U_{out} = U_Z$; **Output current:** $I_{out} = U_Z/R_L$;

Current: $I = I_{out} + I_Z = U_Z/R_L + I_Z$;

$$\because U_{in} = U_Z + IR; \quad \therefore R = (U_{in} - U_Z)/I = \frac{U_{in} - U_Z}{U_Z/R_L + I_Z}$$

$$\therefore \frac{U_{in} - U_Z}{U_Z/R_L + I_{ZMAX}} < R < \frac{U_{in} - U_Z}{U_Z/R_L + I_{ZMIN}}$$

Homework 1: please refer to QQ Group and blackboard. (暂时不提交，和下次一起)